

Optimal Public Debt Adjustment with Heterogeneous Agents: The Role of Aggregate Risk

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September 2026

Preliminary draft

Abstract

How fast should a government repay the debt it accumulates in a recession? I study this question in a heterogeneous-agent economy with capital, uninsurable earnings risk, borrowing constraints, and aggregate productivity shocks, in which the government repays debt by raising labor income taxes. Public debt crowds out capital and lowers wages, a cost borne disproportionately by hand-to-mouth households, and it raises interest rates, which benefits wealthy households. I characterize the optimal debt-adjustment rule and show that, with aggregate risk, it depends on how these social costs comove with the level of debt. Quantitatively, the welfare-maximizing adjustment coefficient is more than ten times larger with heterogeneous households than with a representative agent, and the implied persistence of public debt is close to that observed in the United States. Neither ingredient suffices on its own: without aggregate risk, the planner prefers to repay as slowly as possible. Optimal policy therefore departs sharply from tax smoothing.

JEL codes: E32, E62, E64, H63.

Keywords: Optimal fiscal policy, public debt, fiscal rules, heterogeneous-agent models, aggregate risk, tax smoothing, redistribution.

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I am grateful to Fabrice Collard and Andreas Schaab for their invaluable advice and continuous guidance. I also thank Mikhail Golosov, Eugenia Gonzalez-Aguado, Alexandre Gaillard, Christian Hellwig, Patrick Kehoe, Narayana Kocherlakota, Elena Pastorino, and participants at the SAEe 2024 conference, the TSE Macro workshop, and the TSE–Bank of Mauritius workshop for helpful comments.

1 Introduction

Fiscal rules require governments to repay the debt they accumulate. The Stability and Growth Pact limits the debt and deficits of the members of the European Union and requires those above the limits to reduce their debt at a minimum pace.¹ Many member states miss these targets.² The European Commission’s own assessment summarizes the trade-off. High debt places countries at a “*high risk over the medium term*”, yet “*compliance with the debt reduction benchmark could imply a too demanding frontloaded fiscal effort that could risk to jeopardise economic growth*”. Neither the cost of repaying debt nor the cost of carrying it is shared equally across households. This paper asks how quickly a government should repay the debt it accumulates during a recession, and how the answer depends on household heterogeneity and aggregate risk.

With a representative agent, the tax-smoothing prescription of Barro (1979) and Lucas and Stokey (1983) applies: the government lets debt absorb shocks so that distortionary taxes need not move much. Household heterogeneity changes the costs of carrying debt. With productive capital, public debt crowds out capital, which raises the interest rate and lowers wages. Households with little wealth, whose labor earnings are their main source of income, bear the wage loss, while the higher interest income accrues to wealthy households. Repaying debt quickly is costly too: it requires higher labor income taxes today, which directly lower the consumption of hand-to-mouth households. Aggregate risk changes the timing of these costs. Without it, a government can wait for debt to fall as the economy recovers. With it, a new recession can arrive while debt is still high and push it higher. Do household heterogeneity and aggregate risk justify repaying debt faster?

I answer this question in a continuous-time heterogeneous-agent model with capital, uninsurable earnings risk, a borrowing constraint, and aggregate TFP shocks, building on Aiyagari (1994) and Krusell and Smith (1998). The government finances constant spending with a labor income tax and risk-free debt. The tax rate responds to deviations of debt from its long-run level through a debt-adjustment coefficient τ^a , which governs how fast debt is repaid. A utilitarian Ramsey planner chooses τ^a , taking into account how households’ consumption, labor supply, and savings respond. As in the optimal-

¹ Countries whose governments exceed a ratio of debt-to-GDP of 60 percent or a deficit-to-GDP ratio of 3 percent are required to undergo a series of fiscal adjustments, with a minimum annual adjustment of at least 0.5 percent of GDP (Darvas and Zettelmeyer, 2023). The 2024 reform of the EU economic governance framework replaced the debt-reduction benchmark with country-specific net expenditure paths, but it retains the requirement that debt above 60 percent of GDP be put on a declining path.

² See the 2023 report of the European Commission, <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52023DC0631>. Due to the economic conditions in 2022, “*compliance with the debt reduction benchmark is not warranted*”.

simple-rules approach of [Schmitt-Grohé and Uribe \(2007\)](#), the planner optimizes over the coefficient of a policy rule rather than over the unrestricted set of fiscal instruments. Households forecast aggregate capital and debt with perceived laws of motion in the spirit of [Krusell and Smith \(1998\)](#), which keeps the planner’s problem tractable in the presence of aggregate risk.³ I derive the planner’s optimality conditions with and without aggregate risk, and I solve the model globally.

The costs of public debt. I first characterize optimal debt adjustment without aggregate risk. Writing the planner’s Lagrangian as in [Dávila and Schaab \(2023\)](#), I show that the optimal coefficient equals excess debt, relative to its long-run level, times the sum of the social costs of debt. These costs fall into three groups. A representative-agent component captures the crowding out of capital. An incomplete-markets component captures the capital income that households gain when debt raises the interest rate, which argues for slower repayment. A third group captures inequality concerns: capital income gains accrue to wealthy households, and hand-to-mouth households lose consumption when debt or taxes rise. In the representative-agent limit, optimal policy reduces to a targeting rule in which the social cost of debt accumulates the cost of crowding out.

Aggregate risk and the timing of repayment. With aggregate risk, the level of excess debt no longer determines the optimal coefficient. Instead, the long-run optimal coefficient depends on the covariance, over the business cycle, between government debt and the social costs of debt. If crowding out, the redistributive effects of higher interest rates, or the losses of hand-to-mouth households are larger precisely when debt is high, as they are in a recession and its aftermath, the planner repays faster. Two additional terms arise because the planner’s choice of τ^a changes households’ forecasts of future capital and debt.

How fast to repay. I calibrate the model to the US economy and solve for the welfare-maximizing coefficient. With heterogeneous households and aggregate risk, the optimal coefficient is $\tau^a = 5.85$: a debt-to-GDP ratio one percentage point above its 70 percent target raises the labor income tax from 33 to about 41 percent. In the representative-agent version of the same economy, the optimal coefficient is 0.54, more than ten times smaller, and the same excess debt raises the tax by less than one percentage point. After a negative TFP shock, optimal policy front-loads repayment. The tax increase on impact is about twice as

³ With aggregate uncertainty, rational expectations are not tractable because the distribution of wealth, an infinite-dimensional object, enters the state space: households need to keep track of the whole distribution of wealth and income to predict tomorrow’s prices. [Moll \(2024\)](#) discusses this and other avenues to tackle the challenge of rational expectations in heterogeneous-agent economies.

large as with a representative agent, debt rises far less, and most of the excess debt is repaid within four years, while consumption takes more than two decades to recover. The implied annual persistence of government debt is 0.91, close to the 0.93 estimated for the United States by [Auclert and Rognlie \(2018\)](#). Both ingredients are necessary. Without aggregate risk, welfare decreases in τ^a in both the heterogeneous-agent and the representative-agent economies, so the planner prefers to repay as slowly as possible. Finally, when households are myopic about the future path of aggregates, the optimal coefficient more than triples, to 19.95. The anticipatory response of savings to future taxes is therefore a first-order cost of fast repayment.

Related literature. This paper contributes to the literature on optimal fiscal policy. [Barro \(1979\)](#) and [Lucas and Stokey \(1983\)](#) establish the tax-smoothing result: the government insures against unexpected shocks, such as a war that temporarily raises government spending, by adjusting public debt, and it keeps labor income taxes smooth. [Lucas and Stokey \(1983\)](#) derive this result with a representative agent and state-contingent debt, and [Aiyagari et al. \(2002\)](#) show that it largely survives when the government can issue only risk-free debt. [Angeletos et al. \(2023a\)](#) study an economy in which public debt provides liquidity and find that, at the optimum, the response of taxes to a government spending shock is front-loaded; as in my economy, their result depends on the premium households pay to hold debt, the gap between the discount rate and the interest rate. [Bhandari et al. \(2017b\)](#) introduce aggregate risk and find that the optimal level of debt is close to zero and that debt mean-reverts very slowly, with a half-life of hundreds of years, in an economy without binding borrowing constraints, and [Bhandari et al. \(2017a\)](#) study heterogeneous households and incomplete markets without aggregate shocks. Relative to this work, I study a smaller set of instruments, which allows me to analyze heterogeneous households, binding borrowing constraints, and aggregate risk together. The optimal rule departs sharply from tax smoothing, and the departure arises only when heterogeneous households face aggregate risk.

The paper also relates to optimal fiscal policy in the Bewley–Huggett–Aiyagari tradition ([Bewley, 1983](#); [Huggett, 1993](#); [Aiyagari, 1994](#)). Much is known about the positive consequences of public debt with household heterogeneity,⁴ and the normative side has received growing attention. [Aiyagari \(1995\)](#) show that optimal capital taxes are positive, assuming a Ramsey steady state exists, and [Aiyagari and McGrattan \(1998\)](#) compute the optimal long-run level of public debt. Recent work studies whether a Ramsey steady state exists. [Chien and Wen \(2022\)](#) and [Auclert et al. \(2023\)](#) find that it does not with separable

⁴ See, among many others, [Heathcote \(2005\)](#), [Auclert et al. \(2018\)](#), and [Ferriere and Navarro \(2023\)](#).

CRRA preferences: the planner wants to accumulate debt to close the gap between the interest rate and the discount rate. [LeGrand and Ragot \(2023\)](#) establish existence for a class of preferences, and [Bayas-Erazo \(2023\)](#) use heterogeneous welfare weights, as in [Dávila and Schaab \(2024\)](#), to restore it. I focus instead on the speed at which debt returns to its long-run level after a shock, abstract from capital income taxes,⁵ and introduce aggregate risk. My results connect to the existence debate. Without aggregate risk, welfare decreases in the debt-adjustment coefficient, echoing the planner’s desire to hold debt in these papers. With aggregate risk, welfare has an interior maximum: aggregate risk gives the planner a reason to repay.

On the timing of fiscal adjustment, [Angeletos et al. \(2023b\)](#) argue that delaying adjustment can be optimal. A deficit-financed boom is persistent, debt returns towards trend on its own, and the required future tax increase shrinks. Their economy features uninsurable income risk but no aggregate risk. With aggregate risk, a new recession may arrive while debt is still high, which weakens the case for waiting.

Methodologically, I extend the Lagrangian approach of [Dávila and Schaab \(2023\)](#), developed for optimal monetary policy without aggregate risk, to a fiscal problem with aggregate shocks and perceived laws of motion. The approach characterizes the social costs of public debt in terms of multipliers with an economic interpretation, including the utility costs of hand-to-mouth households at the borrowing constraint. [Ahn et al. \(2018\)](#) extend linearization techniques to heterogeneous-agent economies with aggregate risk, and [Dávila and Schaab \(2023\)](#) use sequence-space Hessians, which generalize the sequence-space Jacobians of [Auclert et al. \(2021\)](#) and build on [Boppart et al. \(2018\)](#), to evaluate welfare to second order. I instead solve the model globally, to capture the nonlinearities induced by borrowing constraints and aggregate risk.

My contribution relative to this work is threefold. First, I characterize the optimal speed of debt repayment with heterogeneous households and aggregate risk, and show that it depends on how the social costs of debt comove with the level of debt over the business cycle. Second, I show quantitatively that household heterogeneity and aggregate risk are jointly necessary for fast repayment: each ingredient alone calls for slow or no repayment, whereas together they call for a front-loaded fiscal adjustment with debt dynamics close to those in US data. Third, I extend the Lagrangian characterization of optimal policy in heterogeneous-agent economies to aggregate risk, where the planner internalizes how policy shapes households’ forecasts of aggregate variables.

⁵ Capital income taxes are the topic of [Straub and Werning \(2020\)](#) and [LeGrand and Ragot \(2023\)](#), who revisit the [Chamley \(1986\)](#)–[Judd \(1985\)](#) result in a richer setting.

Layout. Section 2 introduces the model. Section 3 derives optimal debt adjustment without aggregate shocks, and Section 4 does so with aggregate risk. Section 5 contains the quantitative assessment, and Section 6 concludes. Proofs are in Appendix A, and Appendix B describes the numerical algorithm.

2 Model

The model builds on the heterogeneous-agent production economy of Aiyagari (1994), with elastic labor supply and a government that finances spending with labor income taxes and debt. When I analyze the role of aggregate risk, TFP follows a stochastic process, as in Krusell and Smith (1998); otherwise, I consider one-time unanticipated TFP shocks. Relative to the textbook real business cycle model (Kydland and Prescott, 1982; Romer, 2012), households face uninsurable labor income risk.

2.1 Households

The economy is populated by a continuum of households of mass one. Facing idiosyncratic earnings risk and, when relevant, aggregate uncertainty, households make consumption, labor supply, and savings decisions. The idiosyncratic state of a household consists of its wealth and its earnings status.

Household preferences are defined over consumption and labor,

$$\max \mathbb{E}_0 \int_0^\infty e^{-\rho t} u(c_t, \ell_t) dt, \quad (1)$$

where c_t is the rate of consumption and ℓ_t is the rate of labor supply. Households save in a single liquid, risk-free asset, in quantity a_t , whose return is denoted r_t . Wealth evolves according to

$$\dot{a}_t = r_t a_t + (1 - \tau_t) w_t z_t \ell_t - c_t, \quad (2)$$

where effective units of labor are given by $z_t \ell_t$, and where z_t represents uninsurable labor earnings risk. The wage per effective unit of labor is w_t , and τ_t is a time-varying labor income tax. Households are subject to a borrowing constraint, $a_t \geq \underline{a}$ with $\underline{a} \leq 0$.

Capital and government bonds are perfect substitutes and pay the same return r_t , so that aggregate household wealth equals the sum of the capital stock and government debt.⁶ Households therefore choose how much to save, but not how to allocate their

⁶ In continuous time, with capital that can be converted into consumption one-for-one, the return on capital over an instant, $r_t dt$, is known at t even when TFP is stochastic. No-arbitrage then equates the

savings between the two assets.

Labor earnings risk. Log labor productivity follows a continuous-time AR(1) process, formally an Ornstein–Uhlenbeck process,

$$d \log(z_t) = -\theta_z \log(z_t) dt + \sigma_z dW_t, \quad (3)$$

where $\theta_z > 0$ governs mean reversion, σ_z denotes volatility, and W_t is an individual-specific Brownian motion.

Cross-sectional distribution. Since I abstract from permanent heterogeneity, each household is indexed by a pair of wealth and earnings (a, z) , and the cross-sectional distribution is denoted $g_t(a, z)$. With aggregate risk, I write $g_t(a, z, \Gamma)$ for the joint density over idiosyncratic states and the aggregate state Γ defined below.⁷

2.2 Firms

A representative firm produces the consumption good using capital K_t and labor L_t according to a Cobb–Douglas technology,

$$Y_t = Z_t K_t^\alpha L_t^{1-\alpha},$$

where Z_t is total factor productivity (TFP), the exogenous source of aggregate shocks. Without aggregate uncertainty, I consider deterministic, potentially time-varying paths of TFP $\{Z_t\}_{t=0}^\infty$. With aggregate uncertainty, log TFP follows a continuous-time AR(1) process.⁸

return on capital with the return on short-term government bonds. This is the standard asset structure in continuous-time Krusell–Smith economies (Ahn et al., 2018). An economy in which households choose how to split their wealth between capital and government debt, and in which the two assets differ in their risk and liquidity, is an extension I am currently developing.

⁷ The cross-sectional distribution at a given aggregate state is $g_t(a, z, \Gamma) / g_\Gamma(\Gamma)$, where g_Γ is the marginal density of Γ . Market clearing holds state by state, and cross-sectional aggregates at a given Γ are computed with respect to this conditional distribution.

⁸ Precisely, log productivity follows

$$d \log(Z_t) = -\theta_Z \log(Z_t) dt + \sigma_Z d\tilde{W}_t,$$

where $\tilde{W}_t$ is a Brownian motion common to all households.

Input markets are competitive,

$$w_t = (1 - \alpha) \frac{Y_t}{L_t} \quad \text{and} \quad r_t = \alpha \frac{Y_t}{K_t} - \delta,$$

where δ is the depreciation rate of capital.

2.3 Government

The government uses tax revenue and debt to finance a constant level of government spending $\bar{G}$. Government debt evolves according to

$$\dot{B}_t = r_t B_t + \bar{G} - \tau_t w_t L_t,$$

where B_t denotes short-term risk-free government bonds, which pay the same return as capital by no-arbitrage. The labor income tax is chosen so that government debt mean-reverts to a level $\bar{B} = \mathbb{E}[B_t]$, and the speed of adjustment is governed by the debt-adjustment coefficient τ_t^a ,

$$\tau_t = \bar{\tau} + \tau_t^a (B_t - \bar{B}). \quad (4)$$

The base rate $\bar{\tau}$ balances the budget in the deterministic steady state, denoted with bars: $\bar{r}\bar{B} + \bar{G} = \bar{\tau}\bar{w}\bar{L}$. The coefficient $\tau_t^a > 0$ captures the speed at which public debt is repaid and can be interpreted as the degree of fiscal austerity over the business cycle. Since government spending is constant,⁹ debt above its long-run level can be repaid in three ways: (i) lower interest rates, which reduce debt service; (ii) higher wages, which raise tax revenue at a given tax rate; and (iii) higher tax rates, governed by τ_t^a .

2.4 Equilibrium

I first define a competitive equilibrium without aggregate shocks. I then introduce TFP shocks and specify how households form expectations about future aggregate variables.

Definition 1. (Competitive Equilibrium without Aggregate Shocks) *Given an initial distribution over wealth and earnings, $g_0(a, z)$, a path of the debt-adjustment coefficient $\{\tau_t^a\}$, and a path of TFP $\{Z_t\}$, an equilibrium consists of paths for prices $\{r_t, w_t\}$, aggregates $\{Y_t, L_t, C_t, B_t, K_t, A_t, I_t\}$,*

⁹ I keep government spending constant to focus solely on the adjustment necessary when debt is above its long-run level. Government spending shocks, as an additional source of aggregate risk, would exacerbate the negative consequences of debt whenever a negative TFP shock and a positive spending shock arrive at the same time.

individual policy functions $\{c_t(a, z), \ell_t(a, z), s_t(a, z)\}$, and distributions $\{g_t(a, z)\}$ such that households optimize, firms optimize, and the goods and asset markets clear,

$$Y_t = C_t + I_t + \bar{G}, \quad (5)$$

$$A_t = K_t + B_t, \quad (6)$$

where $C_t = \iint c_t(a, z)g_t(a, z)dadz$, and similarly for the other aggregates, and where $A_t = \iint ag_t(a, z)dadz$ denotes aggregate household wealth. Investment is $I_t = \delta K_t + \dot{K}_t = \delta K_t + \dot{A}_t - \dot{B}_t$, where $\dot{A}_t = \iint s_t(a, z)g_t(a, z)dadz$ is aggregate household saving and $s_t(a, z)$ is individual saving.

With aggregate shocks, the state space is infinite-dimensional. To make both the Ramsey problem and the numerical implementation tractable, I adopt a notion of bounded rationality. All variables are then functions of the aggregate state.

Assumption 2. (Bounded Rationality) Households forecast the future paths of capital and government debt using the current values of TFP, capital, and public debt,

$$\dot{K}_t = \mu_K(\Gamma_t) \quad \text{and} \quad \dot{B}_t = \mu_B(\Gamma_t), \quad (7)$$

where the aggregate state is $\Gamma_t = (Z_t, K_t, B_t)$. These functions minimize the L^2 distance between the predicted and the realized paths of K_t and B_t .

Definition 3. (Competitive Equilibrium with Aggregate Shocks under Assumption 2) Given an initial distribution over wealth, earnings, and the aggregate state, $g_0(a, z, \Gamma)$, and given a debt-adjustment coefficient τ^a , an equilibrium consists of prices $\{r_t(\Gamma), w_t(\Gamma)\}$, aggregates $\{Y_t(\Gamma), L_t(\Gamma), C_t(\Gamma), B_t(\Gamma), K_t(\Gamma), A_t(\Gamma), I_t(\Gamma)\}$, individual policy functions $\{c_t(a, z, \Gamma), \ell_t(a, z, \Gamma), s_t(a, z, \Gamma)\}$, and a distribution $\{g_t(a, z, \Gamma)\}$ such that households optimize, firms optimize, the goods and asset markets clear, and beliefs are formed according to Assumption 2.

2.5 Implementability

The Ramsey planner chooses among allocations that can be implemented as competitive equilibria. I now state the implementability conditions.

First, the planner must respect households' optimal consumption-savings decisions,

$$\partial_c u_t(a, z, \Gamma) = \partial_a V_t(a, z, \Gamma), \quad (8)$$

where $u_t(a, z, \Gamma) = u(c_t(a, z, \Gamma), \ell_t(a, z, \Gamma))$ and $\partial_c u_t(a, z, \Gamma)$ denotes the marginal utility of consumption. This condition equates the marginal utility of consumption to the private marginal value of wealth. Second, the planner must respect the intratemporal trade-off between labor and consumption,

$$\partial_\ell u_t(a, z, \Gamma) = -(1 - \tau_t(\Gamma))w_t(\Gamma)z\partial_c u_t(a, z, \Gamma), \quad (9)$$

which equates the private marginal utility cost of labor, $\partial_\ell u_t(a, z, \Gamma)$, to its private marginal benefit, the utility of consuming the after-tax earnings from working. With aggregate risk, the labor income tax is indexed by the aggregate state Γ . Third, the planner must account for the evolution of private lifetime utility $V_t(a, z, \Gamma)$, given by the Hamilton–Jacobi–Bellman (HJB) equation

$$\rho V_t(a, z, \Gamma) = u_t(a, z, \Gamma) + \mathbb{E}_t \left[\frac{dV_t(a, z, \Gamma)}{dt} \right]. \quad (10)$$

The continuation value is $\mathbb{E}_t \left[\frac{dV_t(a, z, \Gamma)}{dt} \right] = \partial_t V_t(a, z, \Gamma) + \mathcal{A}_t V_t(a, z, \Gamma)$, where $\mathcal{A}_t$ is the infinitesimal generator¹⁰ of the stochastic processes for (a, z, Γ) . Fourth, the planner internalizes that changes in policy affect the mass of households in each state. The cross-sectional distribution evolves according to the Kolmogorov forward equation

$$\partial_t g_t(a, z, \Gamma) = \mathcal{A}_t^* g_t(a, z, \Gamma), \quad (11)$$

where $\mathcal{A}_t^*$ is the adjoint of $\mathcal{A}_t$; see [Achdou et al. \(2022\)](#) for a formal description of these objects. The perceived laws of motion of Assumption 2 are encoded in $\mathcal{A}_t$. Whenever the planner changes policy, she therefore affects not only the evolution of wealth but also households' beliefs about the evolution of capital and government debt. By the choice of the earnings process, I abstract from the cyclicity of idiosyncratic risk.¹¹ Finally, the planner must ensure that the asset market clears and that the government budget

¹⁰ Formally, for any smooth function $f : \mathbb{R}^5 \rightarrow \mathbb{R}$ whose inputs are wealth, earnings, and the aggregate state, the infinitesimal generator is

$$\begin{aligned} \mathcal{A}_t f_t(a, z, \Gamma) = & s_t(a, z, \Gamma) \partial_a f_t(a, z, \Gamma) - \theta_z z \partial_z f_t(a, z, \Gamma) + \frac{\sigma_z^2}{2} \partial_{zz} f_t(a, z, \Gamma) - \theta_Z Z \partial_Z f_t(a, z, \Gamma) + \frac{\sigma_Z^2}{2} \partial_{ZZ} f_t(a, z, \Gamma) \\ & + \mu_K(\Gamma) \partial_K f_t(a, z, \Gamma) + \mu_B(\Gamma) \partial_B f_t(a, z, \Gamma), \end{aligned}$$

where $s_t(a, z, \Gamma)$ denotes savings. The adjoint operator $\mathcal{A}_t^*$ satisfies $\langle \mathcal{A}_t f_1, f_2 \rangle = \langle f_1, \mathcal{A}_t^* f_2 \rangle$ for any two smooth functions $f_1, f_2 : \mathbb{R}^5 \rightarrow \mathbb{R}$, where $\langle \cdot, \cdot \rangle$ is the inner product.

¹¹ [Bilbiie et al. \(2023\)](#) study the quantitative role of countercyclical income risk in a business cycle model, and [Schaab \(2020\)](#) measures its implications for macroeconomic uncertainty.

constraint holds.

3 Optimal Policy without Aggregate Shocks

I begin with the economy without aggregate shocks, which isolates the social costs of public debt that the planner trades off. The state space is (a, z) , and Assumption 2 plays no role. Section 4 reintroduces aggregate risk.¹²

Ramsey problem with commitment. I derive optimal policy with the primal approach, in which the planner chooses among the competitive equilibria that are implementable in the decentralized economy. The planner's only instrument is the debt-adjustment coefficient τ_t^a : she does not choose the base tax rate $\bar{\tau}$ or the long-run debt target $\bar{B}$.

Definition 4. (Ramsey Planner without Aggregate Shocks) A Ramsey planner with infinite commitment horizon $[0, \infty)$ chooses allocations, a policy, and prices¹³

$$\mathbf{X} = \left\{ c_t(a, z), \ell_t(a, z), V_t(a, z), g_t(a, z), \tau_t^a, r_t, w_t, K_t, B_t \right\}_{t=0}^{\infty}$$

as well as multipliers

$$\mathbf{M} = \left\{ \phi_t(a, z), \chi_t(a, z), \psi_t(a, z), \lambda_t(a, z), \mu_t, \theta_t \right\}_{t=0}^{\infty}$$

to maximize social welfare subject to the implementability conditions

$$\rho V_t(a, z) = u_t(a, z) + \partial_t V_t(a, z) + \mathcal{A}_t V_t(a, z) \quad (\text{HJB})$$

$$\partial_c u_t(a, z) = \partial_a V_t(a, z) \quad (\text{FOC } c)$$

$$\partial_\ell u_t(a, z) = -(1 - \tau_t) w_t z \partial_a V_t(a, z) \quad (\text{FOC } \ell)$$

$$\partial_t g_t(a, z) = \mathcal{A}_t^* g_t(a, z) \quad (\text{KFE})$$

$$K_t + B_t = \iint a g_t(a, z) da dz \quad (\text{McK})$$

$$\dot{B}_t = r_t B_t + \bar{G} - \tau_t w_t L_t \quad (\text{GvtBC})$$

¹² While I focus on an infinite commitment horizon $[0, \infty)$, it is possible to study a commitment horizon $[0, \tau_1)$ with $\tau_1 \rightarrow 0$, which I refer to as optimal policy under discretion, as in [Dávila and Schaab \(2023\)](#).

¹³ Relative to the definition of a competitive equilibrium (Definition 1), the Ramsey planner does not optimize directly over individual savings, aggregate savings, and investment: savings follow from the choice of consumption and labor, and goods market clearing follows from Walras' law.

taking as given the initial cross-sectional distribution $g_0(a, z)$, where $\tau_t = \bar{\tau} + \tau_t^a(B_t - \bar{B})$. It solves

$$\mathcal{W} = \min_X \max_M \mathbb{E}_0 [L(g_0)], \quad (12)$$

where the expectation $\mathbb{E}_0$ is taken over idiosyncratic states, and $L(g_0)$ is the planner's Lagrangian given the initial cross-sectional distribution g_0 :

$$\begin{aligned} L(g_0) = \int_0^\infty e^{-\rho t} \Bigg\{ & \iint \left\{ \omega_t(a, z) u_t(a, z) g_t(a, z) \right. \\ & + \phi_t(a, z) \left[-\rho V_t(a, z) + u_t(a, z) + \partial_t V_t(a, z) + \mathcal{A}_t V_t(a, z) \right] \\ & + \chi_t(a, z) \left[\partial_c u_t(a, z) - \partial_a V_t(a, z) \right] \\ & + \psi_t(a, z) \left[\partial_\ell u_t(a, z) + (1 - \tau_t) w_t z \partial_a V_t(a, z) \right] \\ & + \lambda_t(a, z) \left[-\partial_t g_t(a, z) + \mathcal{A}_t^* g_t(a, z) \right] \Bigg\} dadz \\ & + \theta_t \left[-\dot{B}_t + r_t B_t + \bar{G} - \tau_t w_t L_t \right] \\ & + \mu_t \left[\iint a g_t(a, z) dadz - K_t - B_t \right] \Bigg\} dt. \end{aligned} \quad (13)$$

The definition allows for general welfare weights; I focus on the utilitarian case, $\omega(a, z) = 1$. Two of the six implementability conditions, the Bellman equation (**HJB**) and the dynamics of public debt (**GvtBC**), are forward looking. They act as promise-keeping constraints, one on lifetime utility and one on the level of government debt, and they drop out under discretion, when the planner cannot commit to the future path of variables.

Proposition 5. (Necessary Optimality Conditions without Aggregate Shocks) *The first-*

order conditions of the Ramsey planner without aggregate shocks are¹⁴

$$\partial_t \phi_t(a, z) = \mathcal{A}_t^* \phi_t(a, z) + \partial_a \chi_t(a, z) - \partial_a \psi_t(a, z)(1 - \tau_t)w_t z \quad (14)$$

$$\rho \lambda_t(a, z) = u_t(a, z) + \mu_t a - \tau_t w_t \theta_t \ell_t(a, z) + \mathcal{A}_t \lambda_t(a, z) + \partial_t \lambda_t(a, z) \quad (15)$$

$$0 = \partial_c u_t(a, z) - \partial_a \lambda_t(a, z) - \tilde{\chi}_t(a, z) \quad (16)$$

$$0 = \partial_\ell u_t(a, z) + (1 - \tau_t)w_t \partial_a \lambda_t(a, z) - \tilde{\psi}_t(a, z) - \theta_t \tau_t w_t z \quad (17)$$

$$0 = \iint a \left[\phi_t(a, z) \partial_a V_t(a, z) + \partial_a \lambda_t(a, z) g_t(a, z) \right] dadz + \zeta_r^{HtM} + \theta_t B_t \quad (18)$$

$$\begin{aligned} \dot{\theta}_t = & \tau_t^a w_t \iint z \left[\ell_t(a, z) (\phi_t(a, z) \partial_a V_t(a, z) + \partial_a \lambda_t(a, z) g_t(a, z)) + \partial_a \psi_t(a, z) V_t(a, z) \right] dadz \\ & + \tau_t^a \theta_t w_t L_t + \theta_t (\rho - r_t) + \mu_t + \zeta_B^{HtM} \end{aligned} \quad (19)$$

$$\begin{aligned} 0 = & w_t (B_t - \bar{B}) \iint z \left[\ell_t(a, z) (\phi_t(a, z) \partial_a V_t(a, z) + \partial_a \lambda_t(a, z) g_t(a, z)) + \partial_a \psi_t(a, z) V_t(a, z) \right] dadz \\ & + \theta_t (B_t - \bar{B}) w_t L_t + \zeta_\tau^{HtM} \end{aligned} \quad (20)$$

where

$$\begin{aligned} \tilde{\chi}_t(a, z) &= -\frac{\chi_t(a, z)}{g_t(a, z)} \partial_{cc} u_t(a, z) - \frac{\psi_t(a, z)}{g_t(a, z)} \partial_{c\ell} u_t(a, z), \\ \tilde{\psi}_t(a, z) &= -\frac{\psi_t(a, z)}{g_t(a, z)} \partial_{\ell\ell} u_t(a, z) - \frac{\chi_t(a, z)}{g_t(a, z)} \partial_{c\ell} u_t(a, z), \end{aligned}$$

and where ζ_r^{HtM} , ζ_B^{HtM} , and ζ_τ^{HtM} are defined in (31), (32), and (33), respectively. The initial conditions for the forward-looking multipliers are $\theta_0 = \phi_0(a, z) = 0$, and equations (14)–(17) hold in the interior of the state space, because (HJB), (FOC c), (FOC ℓ), and (KFE) are defined only in the interior.

Equations (14) to (19) are the optimality conditions for the value function, the cross-sectional distribution, consumption, labor supply, the interest rate, and government debt; equation (20) is the condition for the debt-adjustment coefficient. Appendix A.1 proves the proposition. The structure parallels Dávila and Schaab (2023), who study optimal monetary policy; here the instrument is fiscal.

¹⁴ Proposition 5 states necessary conditions for an interior optimum, if one exists. Auclert et al. (2023) show that in standard heterogeneous-agent economies there is no interior Ramsey steady state: the planner wants as much government debt as possible, to close the gap between r_t and ρ . Consistent with this result, welfare in my calibrated economy without aggregate shocks decreases in a time-invariant coefficient τ^a (Section 5.3). LeGrand and Ragot (2023) provide preferences under which an interior Ramsey steady state exists.

One remark before interpreting these conditions. I hold $\bar{B}$ fixed at the level of debt that balances the budget at the base tax rate $\bar{\tau}$ in the deterministic steady state. **LeGrand and Ragot (2023)** instead choose the welfare weights $\omega_t(a, z)$ so that a calibrated level of debt is optimal in the deterministic steady state. Because the planner chooses the path $\{B_t\}_t$, a Ramsey allocation can still feature a persistently higher long-run level of debt $B_\infty > \bar{B}$, if this is optimal. The steady-state labor income tax is then higher, $\tau_\infty = \bar{\tau} + \tau_\infty^a (B_\infty - \bar{B}) > \bar{\tau}$, and the planner chooses $\tau_\infty^a > 0$ such that the government budget is balanced. I focus below on the terms that distinguish the planner's conditions from the implementability conditions of Section 2.5.

Distributional penalty. The multiplier $\phi_t(a, z)$ is the social marginal cost of raising the lifetime utility of a household in state (a, z) : a penalty when positive and a reward when negative. I refer to it as a *distributional penalty*. Its law of motion (14) is a Kolmogorov forward equation augmented by the term $\partial_a \chi_t(a, z) - \partial_a \psi_t(a, z)(1 - \tau_t)w_t z$; without this term, the distributional penalty would evolve like the cross-sectional distribution in (KFE). The extra term creates and destroys penalty mass, much as births and deaths do in a life-cycle model, and it is governed by the social costs of consumption and labor discussed below.

Social lifetime value. The multiplier $\lambda_t(a, z)$ is the social value of adding a household in state (a, z) , and equation (15) is a Bellman equation for this *social lifetime value*. It differs from the private HJB equation by the term $\mu_t a - \tau_t w_t \theta_t \ell_t(a, z)$. The first part values the household's assets: μ_t is the social marginal value of an additional unit of asset supply, which relaxes the asset market clearing condition. The second part is the household's tax payment, expressed in social units through θ_t , the social marginal cost of government debt; whenever debt is socially costly, $\theta_t > 0$, it lowers the social lifetime value.

Consumption and labor. Equations (16) and (17) are the planner's counterparts of households' first-order conditions. A household equates its marginal utility of consumption to the private marginal value of wealth; the planner equates it to the social marginal value of wealth, $\partial_a \lambda_t(a, z)$, plus the wedge $\tilde{\chi}_t(a, z)$, which reflects the curvature of utility in consumption and the substitutability between consumption and labor. When $\chi_t(a, z) > 0$, the planner perceives that the household consumes too much, or equivalently saves too little. Similarly, $\psi_t(a, z)$ is the social value of relaxing the intratemporal condition; when $\psi_t(a, z) > 0$, the household works too much from the planner's point of view. The labor

condition contains one additional term, $\theta_t \tau_t w_t z$: a household that works less pays less tax, which is costly whenever debt is socially costly.

Aggregate conditions. Equation (18) is the optimality condition for the interest rate. A higher interest rate raises the income of households with positive wealth, and the planner values the resulting increase in wealth. It lowers the welfare of hand-to-mouth households, as captured by ζ_r^{HtM} , and it raises the cost of servicing government debt.

Equation (19) governs the multiplier θ_t , the social marginal cost of government debt. Whenever $\theta_t > 0$, an increase in debt lowers welfare, so I refer to θ_t as the *public debt penalty*. The penalty evolves for six reasons. First, debt is repaid with a distortionary labor income tax, whose social cost depends on households' before-tax labor income $w_t z \ell_t(a, z)$. Second, higher taxes lower the private gains from working, which the planner values through $\psi_t(a, z)$. Third, the penalty depends on the tax revenue collected today. These three terms are scaled by τ_t^a , because the coefficient determines how debt is repaid. Fourth, debt is costly only to the extent that the interest rate falls short of the discount rate, through $\rho - r_t$, the premium households pay in equilibrium for the ability to self-insure. Fifth, debt crowds out capital: μ_t is the social marginal cost of excess demand for assets, and when $\mu_t > 0$ the penalty rises. Sixth, hand-to-mouth households dislike debt because, other things equal, it lowers their cash on hand, as captured by ζ_B^{HtM} .

Finally, equation (20) is the optimality condition for the debt-adjustment coefficient. The first three motives above, scaled by excess debt $B_t - \bar{B}$, determine the value of repaying faster, and ζ_τ^{HtM} captures the direct loss that a higher coefficient imposes on hand-to-mouth households by lowering their cash on hand.

Optimal public debt adjustment. Combining the necessary conditions yields the main result of this section, the optimal debt-adjustment coefficient without aggregate shocks. Appendix A.2 contains the proof.

Proposition 6. (Optimal Public Debt Adjustment without Aggregate Shocks) *The optimal debt-adjustment coefficient without aggregate shocks is*

$$\tau_t^a = \frac{B_t - \bar{B}}{\zeta_\tau^{\text{HtM}}} \left(\mu_t - \dot{\theta}_t - \frac{\rho - r_t}{B_t} \mathbb{E}_{g_t} \left[a \partial_c u_t(a, z) (1 + \Theta_t(a, z)) \right] + \zeta_B^{\text{HtM}} \right), \quad (21)$$

where $\Theta_t(a, z) = \frac{\phi_t(a, z)}{g_t(a, z)} - \frac{\tilde{\chi}_t(a, z)}{\partial_c u_t(a, z)}$, and where $\mathbb{E}_{g_t}$ is the expectation operator with respect to the cross-sectional distribution.¹⁵

The formula shows how the planner weighs the benefits and costs of repaying debt with distortionary labor income taxes. At a high level, it reads

$$\tau_t^a = (B_t - \bar{B}) \times \text{“sum of the social costs of government debt”}.$$

The larger the excess of debt over its long-run target $\bar{B}$, the larger the coefficient τ_t^a , provided that debt is socially costly in the first place. This proviso is not innocuous: debt need not be costly in the representative-agent case (see Proposition 7 below), nor in the heterogeneous-agent case with standard preferences (Auclert et al., 2023). Equivalently, for a given excess of debt, the more socially costly debt is, the higher the coefficient, and hence the labor tax τ_t . To study the components of these social costs, I decompose the formula as

$$\begin{aligned} \tau_t^a = (B_t - \bar{B}) & \left(\underbrace{\mu_t - \dot{\theta}_t}_{\text{representative agent}} + \underbrace{(r_t - \rho) \frac{A_t}{B_t} \mathbb{E}_{g_t} \left[\partial_c u_t(a, z) (1 + \Theta_t(a, z)) \right]}_{\text{incomplete markets}} \right) \\ & + \underbrace{\frac{r_t - \rho}{B_t} \text{Cov}_{g_t} \left[a, \partial_c u_t(a, z) (1 + \Theta_t(a, z)) \right] + \zeta_B^{\text{HtM}} \frac{1}{\zeta_\tau^{\text{HtM}}}}_{\text{inequality concerns}}, \end{aligned}$$

where the covariance operator is defined using $\mathbb{E}_{g_t}$.

Representative agent. The first term, $\mu_t - \dot{\theta}_t$, is the only one present in a representative-agent economy. The following proposition gives the targeting rule that the planner follows in the representative-agent limit of the framework.

Proposition 7. (Targeting Rule for Public Debt Adjustment without Aggregate Shocks with a Representative Agent) *In the representative-agent limit, and evaluating the interest-rate wedge at its deterministic steady-state value, $r = \rho$, optimal public debt adjustment is characterized by*

$$\dot{\theta}_t = \mu_t. \tag{22}$$

To understand this targeting rule, consider the tax-smoothing benchmark of Lucas and Stokey (1983). There, the planner smooths taxes completely and public debt is costless,

¹⁵ Formally, the cross-sectional expectation operator is defined as $\mathbb{E}_{g_t}[x_t(a, z)] = \iint x_t(a, z) g_t(a, z) da dz$, with $x_t(a, z) : \mathbb{R}^2 \rightarrow \mathbb{R}$ for all t .

so that $\theta_t = 0$. With a supply side, debt raises the demand for assets and crowds out capital. If this is socially costly, $\mu_t > 0$, debt acquires a positive social cost θ_t , which evolves according to (22). As the cost of excess asset demand vanishes, the penalty converges to a stationary, possibly nonzero, level θ^{ss} ; if instead debt permanently raises the demand for assets, $\mu_\infty > 0$, the penalty diverges. Whenever $\mu_t - \dot{\theta}_t > 0$, the planner raises τ_t^a to avoid these costs.

Incomplete markets. The second term arises because markets are incomplete and households pay a premium for holding assets, $r_t < \rho$. Consider first the case $\Theta_t(a, z) = 0$. Since there is a unit mass of households, the average household holds A_t assets, and a higher interest rate induced by more debt raises its capital income. This benefit of debt enters with a negative sign, since $r_t - \rho < 0$, and argues for slower repayment. It is scaled by $1/B_t$, reflecting the cost of servicing a larger stock of debt, and by the gap $r_t - \rho$, since debt is costly only to the extent that the market rate falls short of the discount rate. The term $\Theta_t(a, z)$ adjusts these private gains for the planner's assessment of each household: whether it over- or under-consumes, over- or under-works, and has too high or too low a lifetime utility. If the planner perceives a household as under-consuming, for instance, she under-weights its gains from a higher interest rate.

Inequality concerns. Consider again $\Theta_t(a, z) = 0$. Since marginal utility decreases with wealth, $\text{Cov}_{g_t}[a, \partial_c u_t(a, z)] < 0$, and the covariance is larger in absolute value the more unequal the economy. The interest income generated by debt accrues to wealthy households, while the cost of excess debt falls disproportionately on the wealth-poor; a planner averse to inequality therefore repays faster. This redistribution motive is again scaled by $(r_t - \rho)/B_t$, because the effect of debt on the cross-section of households matters only to the extent that $r_t < \rho$. It weakens if $\Theta_t(a, z)$ comoves positively with wealth and strengthens if it comoves negatively.

Two further terms concern hand-to-mouth households. The first, ζ_B^{HtM} , is the utility cost of higher debt for constrained households. It is positive under mild conditions,¹⁶ since more debt lowers their cash on hand, and it argues for faster repayment. The second, ζ_τ^{HtM} , is the utility cost of a higher coefficient for hand-to-mouth households. It is also positive under mild conditions, and it argues for a lower coefficient.

¹⁶ The condition requires that the consumption loss of constrained households from a higher tax dominates the utility effect of their labor-supply response; otherwise, the sign of ζ_B^{HtM} reverses.

4 Optimal Policy with Aggregate Shocks

I now introduce aggregate risk. The optimal debt-adjustment formula in this environment is the main theoretical result of the paper.

Ramsey problem with commitment. Assumption 2 is now relevant, and the state space is (a, z, Γ) , where $\Gamma = (Z, K, B)$ is the aggregate state. The planner must choose allocations that are consistent with households' beliefs. In particular, for any value $\tilde{B}$ at the aggregate state $\Gamma = (Z, K, \tilde{B})$, the planner chooses a mapping $B_t : \mathbb{R}^3 \rightarrow \mathbb{R}$ such that $B_t(\Gamma) = \tilde{B}$, and the same applies to capital. To lighten notation, I often drop the dependence of government debt and capital on Γ .

Definition 8. (Ramsey Planner with Aggregate Shocks) A Ramsey planner with infinite commitment horizon $[0, \infty)$ chooses allocations, a policy, and prices

$$\mathbf{X} = \left\{ c_t(a, z, \Gamma), \ell_t(a, z, \Gamma), V_t(a, z, \Gamma), g_t(a, z, \Gamma), \tau_t^a, r_t(\Gamma), w_t(\Gamma), K_t(\Gamma), B_t(\Gamma) \right\}_{t=0}^{\infty}$$

as well as multipliers

$$\mathbf{M} = \left\{ \phi_t(a, z, \Gamma), \chi_t(a, z, \Gamma), \psi_t(a, z, \Gamma), \lambda_t(a, z, \Gamma), \mu_t(\Gamma), \theta_t(\Gamma) \right\}_{t=0}^{\infty}$$

to maximize social welfare subject to the implementability conditions

$$\rho V_t(a, z, \Gamma) = u_t(a, z, \Gamma) + \partial_t V_t(a, z, \Gamma) + \mathcal{A}_t V_t(a, z, \Gamma) \quad (\text{HJB}(\Gamma))$$

$$\partial_c u_t(a, z, \Gamma) = \partial_a V_t(a, z, \Gamma) \quad (\text{FOC } c(\Gamma))$$

$$\partial_\ell u_t(a, z, \Gamma) = -(1 - \tau_t) w_t(\Gamma) z \partial_c u_t(a, z, \Gamma) \quad (\text{FOC } \ell(\Gamma))$$

$$\partial_t g_t(a, z, \Gamma) = \mathcal{A}_t^* g_t(a, z, \Gamma) \quad (\text{KFE}(\Gamma))$$

$$K_t + B_t = \iint a g_t(a, z, \Gamma) da dz \quad (\text{McK}(\Gamma))$$

$$\dot{B}_t = r_t(\Gamma) B_t + \bar{G} - \tau_t w_t(\Gamma) L_t(\Gamma) \quad (\text{GvtBC}(\Gamma))$$

taking as given the initial distribution $g_0(a, z, \Gamma)$, where $\tau_t = \bar{\tau} + \tau_t^a (B_t(\Gamma) - \bar{B})$. That is,

$$\mathcal{W} = \min_{\mathbf{X}} \max_{\mathbf{M}} \mathbb{E}_0 [L(g_0)],$$

where the expectation $\mathbb{E}_0$ is taken over idiosyncratic and aggregate states, and $L(g_0)$ is the planner's Lagrangian given the initial distribution g_0 :

$$\begin{aligned}
L(g_0) = \int_0^\infty e^{-\rho t} \int \left\{ \iint \left\{ \omega_t(a, z, \Gamma) u_t(a, z, \Gamma) g_t(a, z, \Gamma) \right. \right. \\
+ \phi_t(a, z, \Gamma) \left[-\rho V_t(a, z, \Gamma) + u_t(a, z, \Gamma) + \partial_t V_t(a, z, \Gamma) + \mathcal{A}_t V_t(a, z, \Gamma) \right] \\
+ \chi_t(a, z, \Gamma) \left[\partial_c u_t(a, z, \Gamma) - \partial_a V_t(a, z, \Gamma) \right] \\
+ \psi_t(a, z, \Gamma) \left[\partial_\ell u_t(a, z, \Gamma) + (1 - \tau_t) w_t(\Gamma) z \partial_a V_t(a, z, \Gamma) \right] \\
+ \lambda_t(a, z, \Gamma) \left[-\partial_t g_t(a, z, \Gamma) + \mathcal{A}_t^* g_t(a, z, \Gamma) \right] \left. \right\} dadz \\
+ \theta_t(\Gamma) \left[-\dot{B}_t + r_t(\Gamma) B_t + \bar{G} - \tau_t w_t(\Gamma) L_t(\Gamma) \right] \\
+ \mu_t(\Gamma) \left[\iint a g_t(a, z, \Gamma) dadz - K_t - B_t \right] \left. \right\} d\Gamma dt.
\end{aligned}$$

As in the previous section, I focus on the utilitarian case, $\omega(a, z, \Gamma) = 1$. The six implementability conditions are ordered as before: the Bellman equation, the first-order conditions for consumption and labor, the evolution of the cross-sectional distribution, market clearing, and the dynamics of debt. All objects are now indexed by the aggregate state Γ .

Proposition 9. (Necessary Optimality Conditions with Aggregate Shocks) *The first-order conditions of the Ramsey planner with aggregate shocks are¹⁷*

$$\partial_t \phi_t(a, z, \Gamma) = \mathcal{A}_t^* \phi_t(a, z, \Gamma) + \partial_a \chi_t(a, z, \Gamma) - \partial_a \psi_t(a, z, \Gamma) (1 - \tau_t) w_t(\Gamma) z \quad (23)$$

$$\rho \lambda_t(a, z, \Gamma) = u_t(a, z, \Gamma) + \mu_t(\Gamma) a - \tau_t w_t(\Gamma) \theta_t(\Gamma) \ell_t(a, z, \Gamma) + \mathcal{A}_t \lambda_t(a, z, \Gamma) + \partial_t \lambda_t(a, z, \Gamma) \quad (24)$$

$$0 = \partial_c u_t(a, z, \Gamma) - \partial_a \lambda_t(a, z, \Gamma) - \tilde{\chi}_t(a, z, \Gamma) \quad (25)$$

¹⁷ Footnote 14 summarizes what is known about the existence of a Ramsey allocation in heterogeneous-agent economies without aggregate shocks. In the economy with aggregate shocks and under the notion of bounded rationality of Assumption 2, I verify numerically that a stochastic steady state exists for a wide range of values of τ^a and that social welfare has an interior maximum in τ^a .

$$0 = \partial_\ell u_t(a, z, \Gamma) + (1 - \tau_t)w_t(\Gamma)\partial_a \lambda_t(a, z, \Gamma) - \tilde{\psi}_t(a, z, \Gamma) - \theta_t(\Gamma)\tau_t w_t(\Gamma)z \quad (26)$$

$$0 = \iint a \left[\phi_t(a, z, \Gamma)\partial_a V_t(a, z, \Gamma) + \partial_a \lambda_t(a, z, \Gamma)g_t(a, z, \Gamma) \right] dadz + \zeta_r^{HtM}(\Gamma) + \theta_t(\Gamma)B_t \quad (27)$$

$$\begin{aligned} \dot{\theta}_t(\Gamma) = & \tau_t^a w_t(\Gamma) \iint z \left[\ell_t(a, z, \Gamma)(\phi_t(a, z, \Gamma)\partial_a V_t(a, z, \Gamma) + \partial_a \lambda_t(a, z, \Gamma)g_t(a, z, \Gamma)) \right. \\ & \left. + \partial_a \psi_t(a, z, \Gamma)V_t(a, z, \Gamma) \right] dadz + \tau_t^a \theta_t(\Gamma)w_t(\Gamma)L_t(\Gamma) + \theta_t(\Gamma)(\rho - r_t(\Gamma)) \\ & + \mu_t(\Gamma) + \zeta_B^{HtM}(\Gamma) - \mathcal{E}_B(\Gamma) \end{aligned} \quad (28)$$

$$\begin{aligned} 0 = & w_t(\Gamma)(B_t - \bar{B}) \iint z \left[\ell_t(a, z, \Gamma)(\phi_t(a, z, \Gamma)\partial_a V_t(a, z, \Gamma) + \partial_a \lambda_t(a, z, \Gamma)g_t(a, z, \Gamma)) \right. \\ & \left. + \partial_a \psi_t(a, z, \Gamma)V_t(a, z, \Gamma) \right] dadz + \theta_t(\Gamma)(B_t - \bar{B})w_t(\Gamma)L_t(\Gamma) + \zeta_\tau^{HtM} + \mathcal{E}_\tau \end{aligned} \quad (29)$$

where

$$\begin{aligned} \tilde{\chi}_t(a, z, \Gamma) &= -\frac{\chi_t(a, z, \Gamma)}{g_t(a, z, \Gamma)}\partial_{cc}u_t(a, z, \Gamma) - \frac{\psi_t(a, z, \Gamma)}{g_t(a, z, \Gamma)}\partial_{c\ell}u_t(a, z, \Gamma), \\ \tilde{\psi}_t(a, z, \Gamma) &= -\frac{\psi_t(a, z, \Gamma)}{g_t(a, z, \Gamma)}\partial_{\ell\ell}u_t(a, z, \Gamma) - \frac{\chi_t(a, z, \Gamma)}{g_t(a, z, \Gamma)}\partial_{c\ell}u_t(a, z, \Gamma), \end{aligned}$$

where $\mathcal{E}_B(\Gamma)$ and $\mathcal{E}_\tau$ are defined in (34) and (35), and $\zeta_r^{HtM}(\Gamma)$, $\zeta_B^{HtM}(\Gamma)$, and ζ_τ^{HtM} are defined in (36), (37), and (38). The initial conditions for the forward-looking multipliers are $\theta_0(\Gamma) = \phi_0(a, z, \Gamma) = 0$, and equations (23)–(26) hold in the interior of the state space, because (HJB(Γ)), (FOC c (Γ)), (FOC ℓ (Γ)), and (KFE(Γ)) are defined only in the interior.

Equations (23) to (28) are the optimality conditions for the value function, the cross-sectional distribution, consumption, labor supply, the interest rate, and government debt; equation (29) is the condition for the debt-adjustment coefficient. Their interpretation is the same as in Section 3, except for the dependence on the aggregate state Γ and the two terms $\mathcal{E}_B(\Gamma)$ and $\mathcal{E}_\tau$. These terms are a byproduct of bounded rationality: they capture the social cost of changing households' beliefs through the planner's actions.

Optimal long-run public debt adjustment. I now extend Proposition 6 to aggregate risk. Hereafter, I focus on the long-run Ramsey allocation.

Proposition 10. (Optimal Public Debt Adjustment with Aggregate Shocks in the Long Run) *The long-run optimal debt-adjustment coefficient with aggregate shocks is*

$$\tau^a = \frac{1}{\zeta_\tau^{\text{HtM}} + \mathcal{E}_\tau} \text{Cov}_\Gamma \left[B, (\rho - r)\tilde{\theta} + \tilde{\mu} + \tilde{\zeta}_B^{\text{HtM}} - \tilde{\mathcal{E}}_B \right],$$

where variables with tildes are normalized by the ergodic distribution of the aggregate state Γ , with associated covariance operator $\text{Cov}_\Gamma[\cdot, \cdot]$.¹⁸

At a high level, the formula reads

$$\tau^a = \text{Cov}_\Gamma \left(B, \text{"sum of the social costs of government debt"} \right).$$

When the social costs of debt comove with the level of debt, that is, when debt is more costly precisely when it is high, the optimal coefficient τ^a is larger. This is the key difference relative to the economy without aggregate shocks. There, what matters is the amount of excess debt; here, what matters is whether debt is more costly in the periods in which it is high.

As in Section 3, I decompose τ^a , dropping the dependence on the state space for convenience,

$$\begin{aligned} \tau^a = \frac{1}{\zeta_\tau^{\text{HtM}} + \mathcal{E}_\tau} & \left(\overbrace{\text{Cov}_\Gamma \left[B, \tilde{\mu} \right] + \text{Cov}_\Gamma \left[B, (r - \rho) \frac{A}{B} \mathbb{E}_{g|\Gamma} \left[u_c(c, \ell) (1 + \Theta) \right] \right]}^{\text{representative agent}} - \text{Cov}_\Gamma \left[B, \tilde{\mathcal{E}}_B \right] \right. \\ & \left. + \underbrace{\text{Cov}_\Gamma \left[B, \frac{r - \rho}{B} \text{Cov}_{g|\Gamma} \left[a, u_c(c, \ell) (1 + \Theta) \right] \right] + \text{Cov}_\Gamma \left[B, \tilde{\zeta}_B^{\text{HtM}} \right]}_{\text{inequality concerns}} \right), \end{aligned}$$

where $g|\Gamma$ is the distribution over idiosyncratic states at a given aggregate state Γ , and $\mathbb{E}_{g|\Gamma}$ and $\text{Cov}_{g|\Gamma}$ are the associated expectation and covariance operators.¹⁹

Representative agent. I start with the representative-agent terms.²⁰

¹⁸ Formally, in the stochastic steady state, TFP shocks move capital and public debt, and a distribution g_Γ emerges over the aggregate state space (TFP, capital, and government debt). This ergodic distribution defines the expectation and covariance operators.

¹⁹ Formally, if $g : \mathbb{R}^5 \rightarrow \mathbb{R}$ is the stochastic steady-state distribution over (a, z, Γ) and g_Γ is the stochastic steady-state distribution of Γ , the distribution over idiosyncratic states (a, z) conditional on the aggregate state Γ is $g_{\text{idio}|\Gamma}(a, z) = \frac{g(a, z, \Gamma)}{g_\Gamma(\Gamma)}$ for all Γ .

²⁰ I assume that $\zeta_\tau^{\text{HtM}} + \mathcal{E}_\tau > 0$.

Corollary 11. (Optimal Public Debt Adjustment in the Representative-Agent Limit) *In the representative-agent limit, the long-run optimal debt-adjustment coefficient is*

$$\tau^a = \frac{1}{\mathcal{E}_\tau} \left(\text{Cov}_\Gamma \left[B, \tilde{\mu} \right] + \text{Cov}_\Gamma \left[B, (r - \rho) \frac{A}{B} \left(u_c(C, L) (1 + \Theta) \right) \right] - \text{Cov}_\Gamma \left[B, \tilde{\mathcal{E}}_B \right] \right).$$

Three terms are present in this formula. First, the social cost of excess asset demand, which captures the crowding out of capital by debt, matters only to the extent that it comoves with the level of debt. It calls for faster repayment if crowding out is more costly when debt is high, that is, in a recession and its aftermath.

Second, the optimal coefficient depends on the comovement of debt with the capital income gains of the average household. A useful approximation of this term is

$$\frac{1}{B} \text{Cov}_\Gamma \left[B, (r - \rho) A \mathbb{E}_{g|\Gamma} \left[u_c(c, \ell) (1 + \Theta) \right] \right].$$

Since $r - \rho < 0$, these capital income gains are a benefit of debt. If they are larger when debt is high, which is when output is low, since debt is countercyclical, the covariance is negative and the planner repays more slowly.

Third, the terms $\tilde{\mathcal{E}}_B$ and $\mathcal{E}_\tau$ capture the social cost of changing households' perceived laws of motion through adjustments in public debt and in the coefficient. These terms disappear when households do not form expectations about the future, that is, when households are myopic. The following corollary gives the optimal coefficient in that case, and Section 5.3 quantifies the role of these terms.

Corollary 12. (Optimal Public Debt Adjustment with Myopic Households) *With myopic households, the long-run optimal debt-adjustment coefficient is*

$$\begin{aligned} \tau^a = \frac{1}{\zeta_\tau^{\text{HtM}}} & \left(\text{Cov}_\Gamma \left[B, (r - \rho) \frac{A}{B} \mathbb{E}_{g|\Gamma} \left[u_c(c, \ell) (1 + \Theta) \right] \right] + \text{Cov}_\Gamma \left[B, \tilde{\mu} \right] \right. \\ & \left. + \text{Cov}_\Gamma \left[B, \frac{r - \rho}{B} \text{Cov}_{g|\Gamma} \left[a, u_c(c, \ell) (1 + \Theta) \right] \right] + \text{Cov}_\Gamma \left[B, \tilde{\zeta}_B^{\text{HtM}} \right] \right). \end{aligned}$$

Inequality concerns. The optimal coefficient also accounts for its effects on the cross-sectional distribution. First, as in Proposition 6, a higher utility cost of the coefficient for hand-to-mouth households, ζ_τ^{HtM} , lowers the optimal coefficient.

Second, the redistribution motive of the economy without aggregate shocks is still

present, but what matters now is how it comoves with the level of debt. To see this, consider the approximation

$$\text{Cov}_\Gamma \left[B, \frac{r - \rho}{B} \text{Cov}_{g|\Gamma} \left[a, u_c(c, \ell) (1 + \Theta) \right] \right] \approx \frac{1}{B} \text{Cov}_\Gamma \left[B, (r - \rho) \text{Cov}_{g|\Gamma} \left[a, u_c(c, \ell) (1 + \Theta) \right] \right].$$

With $\Theta = 0$, both $r - \rho$ and the cross-sectional covariance are negative, so the inner term is positive. If the redistribution motive strengthens in bad times, which is plausible because the consumption of wealth-poor households fluctuates more over the business cycle than that of wealthy households, the inner term is larger when debt is high, the covariance with B is positive, and the planner repays faster. The comovement of $\Theta(a, z, \Gamma)$ may affect this conclusion, which remains a quantitative question.

Third, the comovement of the utility cost of debt for hand-to-mouth households, $\tilde{\zeta}_B^{\text{HtM}}$, with the level of debt raises the optimal coefficient. If an extra unit of debt is more costly for these households when debt is already high, the planner avoids such states by repaying faster.

The utility costs of hand-to-mouth households and the redistribution motive both reflect inequality concerns, but they differ in one respect. The redistribution motive, like the capital income gains discussed above, arises because debt raises the interest rate, and it matters only to the extent that the planner can commit to a future path of government debt. Under discretion, it disappears.

Corollary 13. (*Optimal Public Debt Adjustment under Discretion*) *Under discretion, the long-run optimal debt-adjustment coefficient is*

$$\tau^a = \frac{1}{\zeta_\tau^{\text{HtM}} + \mathcal{E}_\tau} \left(\text{Cov}_\Gamma \left[B, \tilde{\mu} \right] + \text{Cov}_\Gamma \left[B, \tilde{\zeta}_B^{\text{HtM}} \right] - \text{Cov}_\Gamma \left[B, \tilde{\mathcal{E}}_B \right] \right).$$

Under discretion, the trade-off reduces to the losses of hand-to-mouth households from lower after-tax income, the terms from belief formation ($\tilde{\mathcal{E}}_B$ and $\mathcal{E}_\tau$), and the social cost of excess asset demand. All of these costs affect households contemporaneously, either by lowering the cash on hand of constrained households or by crowding out capital and hence lowering wages.

5 Quantitative Assessment

This section computes the welfare-maximizing debt-adjustment coefficient in the economy with heterogeneous agents and aggregate risk, the object characterized in Proposition 10. For comparison, I solve three other versions of the model: with aggregate risk and a representative agent, and without aggregate risk, with heterogeneous agents and with a representative agent. I then study the response of the economy to a negative TFP shock under the optimal rule. In the numerical application, I use a log version of the tax rule (4),

$$\tau_t = \bar{\tau} + \tau^a \ln \left(\frac{B_t}{\bar{B}} \right) \approx \bar{\tau} + \frac{\tau^a}{\bar{B}} (B_t - \bar{B}). \quad (30)$$

To first order, the two rules coincide once the coefficient is rescaled by the constant $1/\bar{B}$. The linear rule (4) is convenient for deriving an explicit formula, while the log specification, also used by McKay and Reis (2016), is better suited for numerical work.

5.1 Calibration

The calibration follows standard parametrizations of one-asset heterogeneous-agent models with capital and government bonds, in particular the “one-account heterogeneous-agent model with high liquidity” of Auclert et al. (2018). The model is calibrated at annual frequency. Table 1 summarizes the parameters.

Households. Preferences are separable between consumption and labor,

$$u(c, \ell) = \frac{c^{1-\sigma}}{1-\sigma} - \Psi \frac{\ell^{1+\nu}}{1+\nu}.$$

In line with Kaplan et al. (2018) and Auclert et al. (2018), I set $\sigma = 1$, which corresponds to log utility in consumption, and $\nu = 1$, a unit Frisch elasticity of labor supply, within the range of estimates in Chetty (2012). I use the scale of the disutility of labor, Ψ , to normalize steady-state output to $Y = 1$. I set the discount rate to $\rho = 0.0658$ to target a steady-state real interest rate of $r = 0.05$, the average combined real return on capital and government bonds from 1969 to 2019. As in Auclert et al. (2018), households cannot borrow, $\underline{a} = 0$, a common assumption in one-asset models in which households hold capital. I calibrate the earnings process to the estimates of the US wage process of Floden and Lindé (2001). The estimated annual persistence of log wages is 0.91, whose continuous-time counterpart is $\theta_z = -\ln(0.91) = 0.0943$. I set the volatility to $\sigma_z = 0.2265$, based on their estimated

Description	Parameter	Value	Target
<i>Preferences</i>			
Discount rate	ρ	0.0658	Real interest rate 5%
Inverse intertemporal elasticity	σ	1	Standard
Inverse Frisch elasticity	ν	1	Standard
<i>Households</i>			
Borrowing constraint	$\underline{a}$	0	Standard
Mean reversion of earnings	θ_z	0.0943	Persistence of log wages
Volatility of earnings	σ_z	0.2265	Volatility of log wages
<i>Firms</i>			
Capital share	α	0.294	Labor share 70.6%
Capital depreciation	δ	Internal	$K/Y = 2.26$
Mean reversion of TFP	θ_Z	1	Autocorrelation of TFP
Volatility of TFP	σ_Z	0.007	Volatility of TFP
<i>Government</i>			
Government spending	$\bar{G}$	0.2	$G/Y = 20\%$
Government debt	$\bar{B}$	0.7	$B/Y = 70\%$
Base tax rate	$\bar{\tau}$	33%	Balanced budget

Table 1. Calibrated Parameters

innovation variance of 0.0426.

Firms. I normalize steady-state TFP to $Z = 1$ and set the labor share to $1 - \alpha = 0.706$. I calibrate the depreciation rate δ internally to target a capital-to-output ratio of $K/Y = 2.26$. To match the autocorrelation and volatility of TFP in the data (Ahn et al., 2018), I set the mean-reversion parameter of log TFP to $\theta_Z = 1$ and its volatility to $\sigma_Z = 0.007$.

Fiscal policy. In the steady state, I target a government spending-to-GDP ratio of $G/Y = 0.2$ and a debt-to-GDP ratio of $B/Y = 0.7$, standard values also used by Auclert et al. (2018) and Campos et al. (2024). The implied base labor income tax rate is $\bar{\tau} = 0.33$.

Implied moments. The calibration implies a ratio of household wealth to pre-tax labor income of $A/(wL) = 4.2$, in line with Kaplan and Violante (2022), and a share of hand-to-mouth households of 30 percent, in line with Kaplan et al. (2018).

5.2 Numerical Implementation

I solve the model globally using the finite-difference method of [Achdou et al. \(2022\)](#), with the sparse-grid tools of [Schaab and Zhang \(2022\)](#) to handle the five-dimensional state space (a, z, Z, K, B) . Following Assumption 2, I compute the stochastic steady state with the [Krusell and Smith \(1998\)](#) algorithm, extended to continuous time as in [Fernández-Villaverde et al. \(2023\)](#). Perceived laws of motion that are linear in the aggregate state deliver a [Den Haan \(2010\)](#) error between perceived and simulated aggregates, the accuracy metric preferred by [Ahn et al. \(2018\)](#), of about 0.2 percent, in line with a baseline Krusell–Smith economy. For a given coefficient τ^a , I compute the stochastic steady state and evaluate utilitarian welfare $\mathcal{W}$ by Monte Carlo simulation. I then search over τ^a for the welfare-maximizing coefficient. Appendix B describes the algorithm in detail.

5.3 Optimal Public Debt Adjustment

I first study the economies with aggregate risk, with heterogeneous agents and with a representative agent. I then switch off aggregate risk and, finally, study the role of households’ expectations about the path of aggregates.

5.3.1 With Aggregate Risk

The welfare-maximizing coefficient in the economy with heterogeneous agents and aggregate risk is $\tau^a = 5.85$, an interior optimum. Under the tax rule (30), a debt-to-GDP ratio one percentage point above its 70 percent target raises the labor income tax from 33 percent to about 41 percent, since $0.33 + 5.85 \ln(0.71/0.70) \approx 0.41$. When debt is at its long-run level, the tax is 33 percent. The optimal rule therefore prescribes strong fiscal consolidation whenever debt is high, including in the early phase of a recession, when the economy is still depressed. This response may seem large relative to the debt-to-GDP ratios observed in practice. At the optimum, however, debt never drifts far from its target, precisely because the adjustment is strong.²¹

The optimal rule implies debt dynamics close to those observed in the data. [Auerclert and Rognlie \(2018\)](#) estimate an autoregressive process for the first difference of US government debt, controlling for changes in government spending, and find an annual

²¹ Actual fiscal systems use several instruments to conduct large fiscal adjustments. For instance, between 2008 and 2014, Greece, Italy, Portugal, and Spain all raised their VAT rates as part of their fiscal adjustments, and they also reformed labor markets, the public sector, and social security. By concentrating the adjustment in a single instrument, the model generates a large optimal labor income tax response, but it isolates the trade-offs that determine the optimal speed of repayment. Additional instruments would plausibly dampen the optimal labor income tax response; I leave them for future research.

persistence of $\rho_B = 0.93$. Estimating the same process on data simulated at the optimum yields $\rho_B^{sim} = 0.91$. The welfare-maximizing rule thus implies debt dynamics similar to those in the United States, with slightly faster repayment. The estimates of [Galí et al. \(2007\)](#), transformed into first differences, imply an annual persistence of 0.903, again close to the optimum.

A representative agent facing the same aggregate risk calls for much slower repayment. In that economy, the optimal coefficient is $\tau^a|^{RA} = 0.54$, more than ten times smaller. The same one-percentage-point excess of debt raises the labor income tax to 33.7 percent, since $0.33 + 0.54 \ln(0.71/0.70) \approx 0.337$, only 0.7 percentage points above its long-run level, against about 8 percentage points with heterogeneous households. The persistence of debt in simulated data rises to $\rho_B^{sim|RA} = 0.98$, above both the data and the heterogeneous-agent optimum. Without households at the borrowing constraint, the planner tolerates persistently high debt.

5.3.2 Without Aggregate Risk

Household heterogeneity alone does not deliver fast repayment. I switch off aggregate risk, so that the only source of risk is idiosyncratic, and hit the economy with a one-time unanticipated TFP shock, which raises debt above its long-run level. In both the heterogeneous-agent and the representative-agent economies, welfare decreases in τ^a over the entire range of coefficients I consider: the planner prefers to repay as slowly as possible, and the optimal coefficient approaches zero. This result mirrors the finding of [Chien and Wen \(2022\)](#) and [Auclert et al. \(2023\)](#) that, without aggregate risk and under standard preferences, a Ramsey planner in a heterogeneous-agent economy wants to accumulate debt to close the gap between the interest rate and the discount rate. Aggregate risk overturns this conclusion, since with it welfare has an interior maximum in τ^a . The two ingredients are therefore jointly necessary: aggregate risk gives the planner a reason to repay, and household heterogeneity makes repayment fast.

5.3.3 The Role of Anticipation

To isolate the role of anticipation, the case of Corollary 12, I set the perceived laws of motion to $\mu_K(\Gamma) = 0$ and $\mu_B(\Gamma) = 0$. Households then expect current prices to persist, and prices change only as TFP moves. The optimal coefficient rises to $\tau^a = 19.95$, close to a complete stabilization of debt: the government offsets any deficit almost immediately with higher taxes. The difference with the baseline measures the social cost of anticipation. When households foresee future tax increases, they save more today to insure against

them, and this response makes fast repayment socially costly. Myopic households do not respond in advance, so the planner can stabilize debt at a lower social cost.

5.4 Optimal Response to a Negative TFP Shock

Figure 1 shows the response to a negative TFP shock in the economy with heterogeneous agents and aggregate risk (HA-AR) and in its representative-agent counterpart (RA-AR), each under its own optimal rule. Because aggregate risk is present, each impulse response averages Monte Carlo simulations in the stochastic steady state that differ only by an additional negative TFP innovation at date 0.

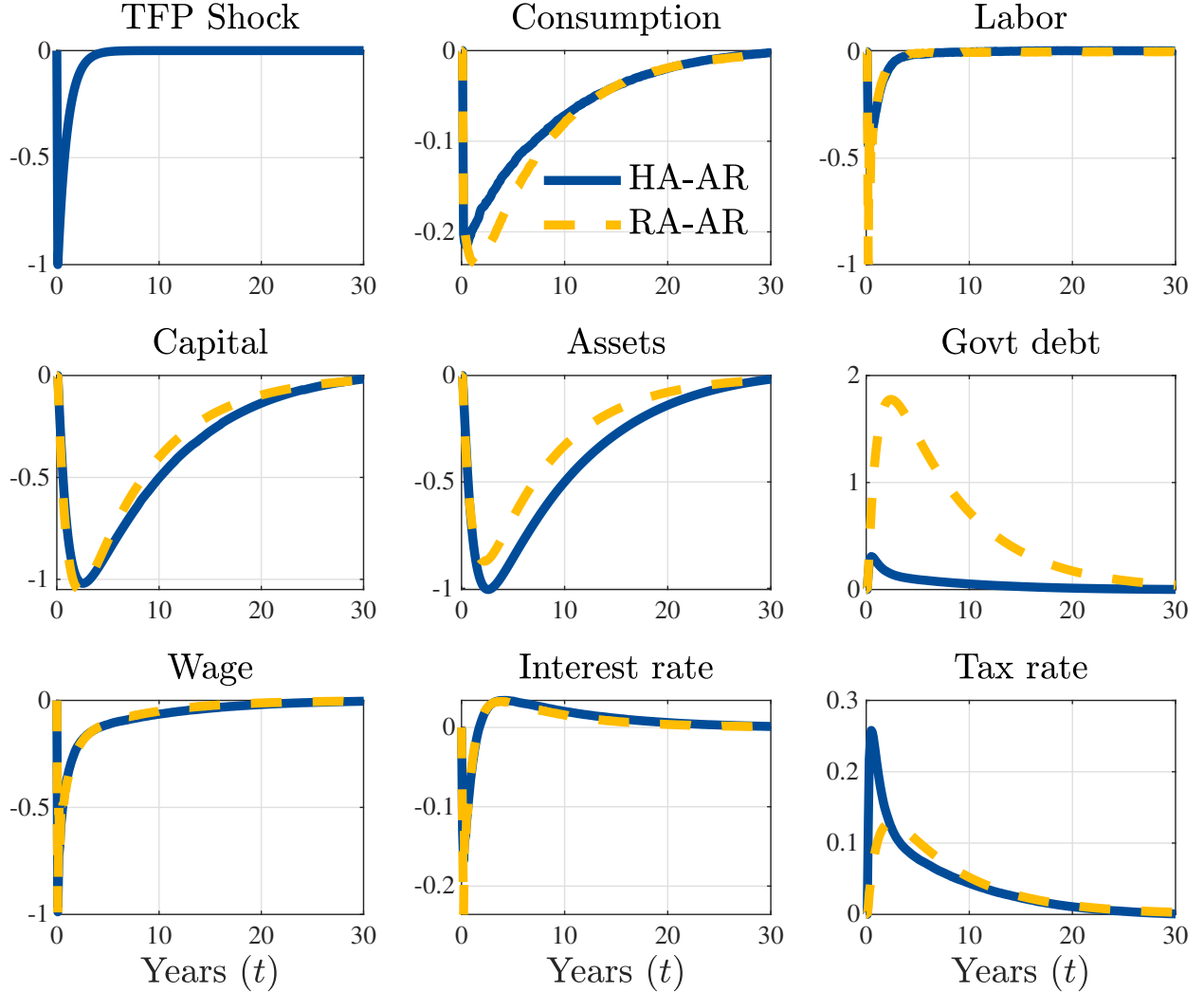
In the heterogeneous-agent economy, aggregates respond as in a standard real business cycle model. Consumption, labor, and capital fall. Labor recovers quickly, while capital, and with it output, recovers slowly. I keep government spending constant, so that the fiscal response isolates the costs of public debt rather than a spending response to the shock. The fall in output therefore raises the ratio of government spending to GDP, and the fall in tax revenue opens a deficit that the government finances by issuing debt. The interest rate falls on impact, with productivity and labor, and then rises above its steady-state level as capital falls.

Optimal policy front-loads the repayment of debt. The labor income tax jumps on impact: a 1 percent decline in TFP raises it by about 0.25 percentage points. Debt rises only briefly, and most of the excess debt is repaid within four years of the shock, whereas consumption needs more than two decades to recover fully. Under a lower coefficient, debt would follow a more pronounced hump-shaped path: during the recession, the tax would not be able to stop debt from rising, and debt would start falling only several years later. After the shock fades, debt again fluctuates around $\bar{B}$.²²

The response with a representative agent is markedly different. The tax rises by about half as much on impact and adjusts gradually. Debt therefore rises by far more, peaking at roughly six times its peak in the heterogeneous-agent economy, and it returns to its long-run level only slowly, consistent with the higher persistence $\rho_B^{sim|RA}$ reported above. Consumption, capital, and wages respond similarly in the two economies. The difference in fiscal policy is therefore not driven by the aggregate response to the shock, but by who bears it. It is consistent with the hand-to-mouth and redistribution terms of Proposition 10,

²² My computational procedure does not determine the optimal deterministic steady-state level of debt around which debt fluctuates; it informs only about the speed of fiscal adjustment. Alternatively, I could follow the inverse-optimal approach of LeGrand and Ragot (2023) and choose the welfare weights $\omega(a, z, \Gamma)$ of Section 4 that make $\bar{B}$ the optimal steady-state level of debt. This approach is unsatisfactory here, because with aggregate risk it may be optimal for debt to fluctuate around a different value than $\bar{B}$.

Figure 1. Impulse Responses to a Negative TFP Shock under the Optimal Rule



Notes: HA-AR is the economy with heterogeneous agents and aggregate risk at its optimal coefficient, $\tau^a = 5.85$; RA-AR is the representative-agent economy with aggregate risk at its optimal coefficient, $\tau^{a|RA} = 0.54$. Each response is the average across Monte Carlo simulations in the stochastic steady state of the difference between paths with and without an additional negative TFP innovation at date 0. Responses are level deviations divided by the size of the innovation to log TFP, so the TFP panel shows a unit shock. Time is in years.

which are absent from Corollary 11.

5.5 Discussion

In Barro (1979) and Lucas and Stokey (1983), the government uses public debt as insurance against unexpected shocks and keeps labor income taxes smooth. The optimal rule in my economy departs sharply from this prescription: after a negative TFP shock, the planner raises taxes on impact and repays debt quickly, as the tax rate panel of Figure 1 shows. My

planner does not choose the long-run level of government debt, so the comparison with [Lucas and Stokey \(1983\)](#) and [Aiyagari et al. \(2002\)](#) is not one-to-one. The comparison across economies within my framework, however, is clean. The departure from tax smoothing arises only when heterogeneous households face aggregate risk: with a representative agent the tax response is small and debt is persistent, and without aggregate risk the planner prefers to repay as slowly as possible.

[Bhandari et al. \(2017b\)](#) study the joint determination of optimal labor income taxes and public debt with aggregate risk. In their economy, without binding borrowing constraints, the government eventually repays its debt, but slowly, with a half-life of at least 250 years depending on the version of the model. The contrast with my results points to the role of borrowing constraints under aggregate risk. When some households cannot self-insure by saving more, the costs of debt fall on households who cannot smooth them, and repayment becomes fast. In the representative-agent economy of [Figure 1](#), where no household is constrained, the fiscal adjustment is much smaller.

Finally, [Angeletos et al. \(2023b\)](#) argue that delaying fiscal adjustment can be optimal: a deficit-financed boom is persistent, debt returns towards trend on its own, and the required future tax increase shrinks. Their result is derived without aggregate risk. With aggregate risk, a government that delays adjustment may be hit by another negative shock while debt is still high, and waiting for debt to fall on its own becomes costly. This again points to the importance of aggregate risk.

6 Conclusion

How quickly should a government repay the debt it accumulates in a recession? I study this question in a heterogeneous-agent economy with capital and aggregate risk, in which a Ramsey planner chooses how strongly labor income taxes respond to excess debt. Carrying debt is costly because it crowds out capital and lowers wages, which hurts households that rely on labor income, while the higher interest rate benefits wealthy households. Repaying debt is costly because it requires higher taxes, which directly hurt hand-to-mouth households. With aggregate risk, the optimal speed of repayment depends on whether these costs are larger when debt is high.

Quantitatively, household heterogeneity and aggregate risk together call for fast repayment. The optimal debt-adjustment coefficient is more than ten times larger than with a representative agent, most of the debt accumulated after a negative TFP shock is repaid within four years, and the implied persistence of debt is close to that observed in the

United States. Without aggregate risk, the planner prefers to repay as slowly as possible. The optimal rule therefore departs sharply from tax smoothing: when heterogeneous households face aggregate risk, excess public debt is too costly to carry.

The analysis leaves several extensions open. First, capital and government bonds are perfect substitutes, so households choose how much to save but not how to split their savings between the two assets. An economy in which households choose a portfolio of capital and government debt, and in which the two assets differ in their risk and liquidity, would let the insurance value of public debt shape the speed of repayment; I am currently developing this extension. Second, the government adjusts a single instrument, the labor income tax. With additional instruments, such as consumption taxes or government spending, the required response of the labor income tax would plausibly be smaller. Third, the planner chooses the coefficient of a fiscal rule. Computing the unrestricted Ramsey policy with aggregate risk, disciplined by the optimality conditions derived in this paper, is a natural next step.

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APPENDIX

A Proofs

A.1 Proof of Proposition 5

I recall the Lagrangian (13), and I drop both time indices and the dependence on the state-space of the economy, namely (a, z) for convenience. Now, I use the index to denote derivatives. It writes,

$$\begin{aligned}
 L = \int_0^\infty e^{-\rho t} \Bigg\{ & \iint \left\{ u(c, \ell) g \right. \\
 & + \phi \left[-\rho V + u(c, \ell) + V_t + \mathcal{A}V \right] \\
 & + \chi \left[u_c(c, \ell) - V_a \right] \\
 & + \psi \left[u_\ell(c, \ell) + (1 - \tau)wzV_a \right] \\
 & + \lambda \left[-g_t + \mathcal{A}^*g \right] \Bigg\} dadz \\
 & + \theta \left[-B_t + rB + G - \tau wL \right] \\
 & + \mu \left[\iint agdadz - K - B \right] \Bigg\} dt
 \end{aligned}$$

where $\tau = \bar{\tau} + \tau^a(B - \bar{B})$. I will first abstract from boundary conditions, and treat them formally later using a “discretize-optimize” approach as in [Dávila and Schaab \(2023\)](#) and [González et al. \(2024\)](#). Integrating by parts, and setting $\phi(0) = \theta(0) = 0$ we get for all

(a, z)

$$\begin{aligned}\int_0^\infty e^{-\rho t} \phi V_t dt &= \rho \int_0^\infty e^{-\rho t} \phi V dt - \int_0^\infty e^{-\rho t} \phi_t V dt \\ \int_0^\infty e^{-\rho t} \lambda g_t dt &= \rho \int_0^\infty e^{-\rho t} \lambda g dt - \int_0^\infty e^{-\rho t} \lambda_t g dt \\ \int_0^\infty e^{-\rho t} \theta B_t dt &= \rho \int_0^\infty e^{-\rho t} \theta B dt - \int_0^\infty e^{-\rho t} \theta_t B dt\end{aligned}$$

Using the definition of the adjoint operator — for any smooth functions f, g , the adjoint operator $\mathcal{A}^*$ of $\mathcal{A}$ satisfies $\langle \mathcal{A}f, g \rangle = \langle f, \mathcal{A}^*g \rangle$, where $\langle \cdot, \cdot \rangle$ is the inner product — we get

$$\int \lambda \mathcal{A}^* g dadz = \int \mathcal{A} \lambda g dadz \quad \text{and} \quad \int \mathcal{A} V \phi dadz = \int V \mathcal{A}^* \phi dadz$$

Finally, integrating by parts yields for all t

$$\int \chi V_a dadz = - \int \chi_a V dadz \quad \text{and} \quad \int (1 - \tau) w z \psi V_a dadz = - \int (1 - \tau) w z \psi_a V dadz$$

where I have abstracted, as mentioned earlier, from boundary conditions. Thus, the Lagrangian rewrites

$$\begin{aligned}L = \int_0^\infty e^{-\rho t} \Big\{ & \iint \left\{ \left[u(c, \ell) + \mu a \right] g \right. \\ & + \phi u(c, \ell) + V \mathcal{A}^* \phi - \phi_t V \\ & + \chi u_c(c, \ell) + \chi_a V \\ & + \psi u_\ell(c, \ell) - \psi_a (1 - \tau) w z V \\ & + \mathcal{A} \lambda g + \lambda_t g - \rho \lambda g \Big\} dadz \\ & + \theta \left[(r - \rho) B + G - \tau w L \right] + \theta_t B \\ & \left. + \mu \left[-K - B \right] \right\} dt\end{aligned}$$

Employing techniques from calculus of variations, consider a perturbation indexed by $\alpha \in \mathbb{R}$, that is, perturb a candidate solution (for example) of the function c in the direction

$c + \alpha h_c$ and so on for the other variables. We have

$$\begin{aligned}
L = \int_0^\infty e^{-\rho t} \Bigg\{ & \iint \left\{ \left[u(c + \alpha h_c, \ell + \alpha h_\ell) + \mu a \right] (g + \alpha h_g) \right. \\
& + \phi u(c + \alpha h_c, \ell + \alpha h_\ell) + (V + \alpha h_V) \mathcal{A}^*(\alpha) \phi - \phi_t (V + \alpha h_V) \\
& + \chi u_c(c + \alpha h_c, \ell + \alpha h_\ell) + \chi_a (V + \alpha h_V) \\
& + \psi u_\ell(c + \alpha h_c, \ell + \alpha h_\ell) - \psi_a \left(1 - \bar{\tau} - (\tau^a + \alpha h_\tau) (B + \alpha h_B - \bar{B}) \right) (w + \alpha h_w) z (V + \alpha h_V) \\
& + \mathcal{A}(\alpha) \lambda (g + \alpha h_g) + \lambda_t (g + \alpha h_g) - \rho \lambda (g + \alpha h_g) \Bigg\} dadz \\
& + \theta \left[(r + \alpha h_r - \rho) (B + \alpha h_B) + G - \left(\bar{\tau} + (\tau^a + \alpha h_\tau) (B + \alpha h_B - \bar{B}) \right) (w + \alpha h_w) \right. \\
& \quad \left. \left(\iint (\ell + \alpha h_\ell) z (g + \alpha h_g) dadz \right) \right] + \theta_t (B + \alpha h_B) \\
& + \mu \left[-K - \alpha h_K - B - \alpha h_B \right] \Bigg\} dt
\end{aligned}$$

Differentiating and taking the limit as $\alpha \rightarrow 0$

$$\begin{aligned}
0 = L_\alpha = \int_0^\infty e^{-\rho t} \Bigg\{ & \iint \left\{ \left[u_c(c, \ell) h_c + u_\ell(c, \ell) h_\ell \right] g + \left[u(c, \ell) + \mu a \right] h_g \right. \\
& + \phi \left[u_c(c, \ell) h_c + u_\ell(c, \ell) h_\ell \right] + \left[\frac{d}{d\alpha} \mathcal{A}^*(0) V + \mathcal{A}^* h_V \right] \phi - \phi_t h_V \\
& + \chi u_{cc}(c, \ell) h_c + \chi u_{c\ell}(c, \ell) h_\ell + \chi_a h_V \\
& + \psi u_{\ell\ell}(c, \ell) h_\ell + \psi u_{\ell c}(c, \ell) h_c - \psi_a (1 - \tau) w z h_V - \psi_a \left((B - \bar{B}) h_\tau + \tau^a h_B \right) w z V \\
& \quad - \psi_a (1 - \tau) h_w z V \\
& + \frac{d}{d\alpha} \mathcal{A}(0) \lambda g + \mathcal{A} \lambda h_g + \lambda_t h_g - \rho \lambda h_g \Bigg\} dadz \\
& + \theta \left[h_r B + (r - \rho) h_B - \left((B - \bar{B}) h_\tau + \tau^a h_B \right) w L - \tau w \left(\iint z (g h_\ell + \ell h_g) dadz \right) \right] + \theta_t h_B \\
& + \mu \left[-h_K - h_B \right] \Bigg\} dt
\end{aligned}$$

where, for smooth functions f, g , we have

$$\iint \frac{d}{d\alpha} \mathcal{A}(0) f g d\alpha dz = \iint \left[ah_r - \left((B - \bar{B})h_\tau + \tau^a h_B \right) w\ell + (1 - \tau)h_w z\ell + (1 - \tau)wh_\ell - h_c \right] f_a g d\alpha dz$$

and it is important to notice that h_τ, h_B , and h_K only depend on the period while the rest of the perturbations are also state-dependent. Now I proceed to apply the fundamental lemma of calculus of variation.

Consumption. Regrouping the terms that are multiplied by h_c

$$0 = u_c(c, \ell)g + \phi u_c(c, \ell) - \phi V_a + \chi u_{cc}(c, \ell) + \psi u_{c\ell}(c, \ell) - \lambda_a g$$

and dividing by g and using the FOC of the competitive equilibrium for consumption

$$0 = u_c(c, \ell) - \lambda_a - \tilde{\chi}; \quad \text{with} \quad \tilde{\chi} = -\frac{\chi}{g} u_{cc}(c, \ell) - \frac{\psi}{g} u_{c\ell}(c, \ell)$$

Labor supply. Regrouping the terms that are multiplied by h_ℓ

$$0 = u_\ell(c, \ell)g + \phi u_\ell(c, \ell) + \phi(1 - \tau)wzV_a + \chi u_{c\ell}(c, \ell) + \psi u_{\ell\ell}(c, \ell) + (1 - \tau)w\lambda_a g - \theta\tau wzg$$

and dividing by g and using the FOC of the competitive equilibrium for the supply of labor

$$0 = u_\ell(c, \ell) + (1 - \tau)w\lambda_a - \tilde{\psi} - \theta\tau wz; \quad \text{with} \quad \tilde{\psi} = -\frac{\psi}{g} u_{\ell\ell}(c, \ell) - \frac{\chi}{g} u_{c\ell}(c, \ell)$$

Lifetime utility. Regrouping the terms that are multiplied by h_V

$$\phi_t = \mathcal{A}^* \phi + \chi_a - \psi_a(1 - \tau)wz$$

Cross-sectional distribution. Regrouping the terms that are multiplied by h_g

$$\rho\lambda = u(c, \ell) + \mu a - \tau w\theta\ell + \mathcal{A}\lambda + \lambda_t$$

Interest Rate. Regrouping the terms that are multiplied by h_r

$$0 = \iint a \left[\phi V_a + \lambda_a g \right] d\alpha dz + \theta B$$

Government debt. Regrouping the terms that are multiplied by h_B

$$0 = -\tau^a w \iint \ell z (\phi V_a + \lambda_a g) dadz - \tau^a w \iint \psi_a z V dadz - \theta \tau^a w L + \theta(r - \rho) - \theta_t - \mu$$

and finally,

$$\theta_t = \tau^a w \iint \left[\ell z (\phi V_a + \lambda_a g) + \psi_a z V \right] dadz + \tau^a \theta w L + \theta(\rho - r) + \mu$$

Debt-adjustment coefficient. Regrouping the terms that are multiplied by h_τ

$$0 = w(B - \bar{B}) \iint \left[\ell z (\phi V_a + \lambda_a g) + \psi_a z V \right] dadz + \theta(B - \bar{B}) w L$$

Accounting for boundary conditions, a “discretize-optimize” approach. The goal now is to formally account for the boundary conditions, that is the fact that the planner cannot freely set consumption for constrained households. Instead, she understands that she can only indirectly affect it by changing the cash-on-hand via the parameters that depend on it.

Now (a, z) are discretized in a matrix in a scattered way, so that row i of the grid corresponds to an individual with wealth a_i and productivity z_i ; with grid size I , and step sizes da and dz . I denote for convenience $dx = dadz$. Grid point $i = 1$ corresponds to the constrained individual, which I assume to be at $(\underline{a}, \underline{z})$; it is straightforward to extend the argument to multiple values of z . The other continuous variable, time, now takes the form of a grid denoted $\{t_n\}_n$, that has size N and time step dt . For convenience, I index a variable $x_t(a, z)$ by x_{in} , and the discretized Lagrangian writes

$$\begin{aligned} L(g_0) = \sum_{n=0}^{N-1} e^{-\rho t_n} \Bigg\{ & \sum_{i=1}^I u(c_{in}, \ell_{in}) g_{in} dx \\ & - \sum_{i>1} \frac{\phi_{in} - \phi_{in-1}}{dt} V_{in} dx + \sum_{i>1} \phi_{in} \left[u(c_{in}, \ell_{in}) + \mathcal{A}_{in} V_{in} \right] dx \\ & + \sum_{i>1} \chi_{in} \left[u_c(c_{in}, \ell_{in}) - \left(\partial_a V \right)_{in} \right] dx \\ & + \sum_{i>1} \psi_{in} \left[u_\ell(c_{in}, \ell_{in}) + (1 - \tau) w z \left(\partial_a V \right)_{in} \right] dx \\ & + \sum_{i>1} \frac{\lambda_{in} - \lambda_{in-1}}{dt} g_{in} dx - \sum_{i>1} \rho \lambda_{in} g_{in} dx + \sum_{i>1} \lambda_{in} \mathcal{A}_{in}^* g_{in} dx \\ & - \mu_n \left[K_n + B_n \right] + \sum_{i=1}^I \mu_n a_i g_{in} dx \end{aligned}$$

$$+ \frac{\theta_n - \theta_{n-1}}{dt} B_n + \theta_n \left[(r_n - \rho) B_n + G_n - \tau w_n \sum_{i=1}^I \ell_{in} z_i \right] \Big\} dt.$$

As pointed out in [Achdou et al. \(2022\)](#), the domain of differentiation for the variable wealth a of the KFE, the HJB, and therefore of the FOC $u_c(c, \ell) = \partial_a V$, is $(\underline{a}, \infty)$. This is why the sum of i 's starts at $i = 2$, and not at $i = 1$. Hence, the necessary conditions for consumption, labor supply, lifetime utility, and cross-sectional distribution only hold in the interior. Having discretized the state-space, now I optimize over r_n , government debt B_n , and the debt-adjustment coefficient τ_n^a . To do so, notice that consumption for the constrained individual writes

$$c_{1n} = r_n a_1 + (1 - \bar{\tau} - \tau_n^a (B_n - \bar{B})) w_n z_1 \ell_{1n}$$

and the supply of labor is given by the intra-temporal substitution of consumption and labor.

Now I can finally provide the necessary conditions including the terms of the borrowing constraint. Formally, I take the FOC directly using the discretized Lagrangian and later take the limit to the continuous state-space

Interest Rate. We have

$$0 = \iint a \left[\phi V_a + \lambda_a g \right] dadz + \zeta_r^{\text{HtM}} + \theta B$$

where

$$\zeta_r^{\text{HtM}} = \left(\underline{a} + (1 - \tau) w z \underline{\ell}_r \right) u_c(\underline{c}, \underline{\ell}) \underline{g} dadz + (1 - \tau) w z \underline{\ell}_r u_\ell(\underline{c}, \underline{\ell}) \underline{g} dadz \quad (31)$$

where $\underline{c}$ (resp. $\underline{\ell}$ with derivative with respect to r denoted $\underline{\ell}_r$) denote consumption (resp. labor supply) of the constrained individual, and where $\underline{g} dadz$ is the measure of individuals at the Dirac mass point $(\underline{a}, \underline{z})$.

Government debt. We have

$$\theta_t = \tau^a w \iint z \left[\ell(\phi V_a + \lambda_a g) + \psi_a V \right] dadz + \tau^a \theta w L + \theta(\rho - r) + \mu + \zeta_B^{\text{HtM}}$$

where

$$\zeta_B^{\text{HtM}} = \left(\tau^a w z \underline{\ell} - (1 - \tau) w z \underline{\ell}_B \right) u_c(\underline{c}, \underline{\ell}) \underline{g} dadz - (1 - \tau) w z \underline{\ell}_B u_\ell(\underline{c}, \underline{\ell}) \underline{g} dadz \quad (32)$$

Debt-adjustment coefficient. And finally, we have

$$0 = w(B - \bar{B}) \iint z \left[\ell(\phi V_a + \lambda_a g) + \psi_a V \right] dadz + \theta(B - \bar{B})wL + \zeta_\tau^{\text{HtM}}$$

where

$$\zeta_\tau^{\text{HtM}} = \left((B - \bar{B})w\underline{z}\underline{\ell} - (1 - \tau)w\underline{z}\underline{\ell}_\tau \right) u_c(\underline{c}, \underline{\ell}) \underline{g} dadz - (1 - \tau)w\underline{z}\underline{\ell}_\tau u_\ell(\underline{c}, \underline{\ell}) \underline{g} dadz \quad (33)$$

which concludes the proof.

A.2 Proof of Proposition 6 and 7

Using

$$\begin{aligned} \theta_t &= \tau^a \left[w \iint z \left[\ell(\phi V_a + \lambda_a g) + \psi_a V \right] dadz + \theta wL \right] + \theta(\rho - r) + \mu + \zeta_B^{\text{HtM}} \\ 0 &= (B - \bar{B}) \left[w \iint z \left[\ell(\phi V_a + \lambda_a g) + \psi_a V \right] dadz + \theta wL \right] + \zeta_\tau^{\text{HtM}} \end{aligned}$$

we get

$$\theta_t = -\tau^a \left[\frac{\zeta_\tau^{\text{HtM}}}{B - \bar{B}} \right] + \theta(\rho - r) + \mu + \zeta_B^{\text{HtM}}$$

and in the RA limit it reduces to $\theta_t = \theta(\rho - r) + \mu$ since $\zeta_\tau^{\text{HtM}} = \zeta_B^{\text{HtM}} = 0$ as there are no hand-to-mouth households in equilibrium. Furthermore, evaluating the interest-rate wedge at its deterministic steady-state value, where $r = \rho$ in the RA limit, we obtain $\theta_t = \mu$. Back in the general case, rearranging,

$$\tau^a = \frac{B - \bar{B}}{\zeta_\tau^{\text{HtM}}} \left(-\theta_t + \theta(\rho - r) + \mu + \zeta_B^{\text{HtM}} \right).$$

We can also use $0 = \iint a \left[\phi V_a + \lambda_a g \right] dadz + \theta B$, to get

$$\tau^a = \frac{B - \bar{B}}{\zeta_\tau^{\text{HtM}}} \left(-\theta_t - \frac{\rho - r}{B} \iint a \left[\phi V_a + \lambda_a g \right] dadz + \mu + \zeta_B^{\text{HtM}} \right)$$

Moreover, since $0 = u_c(c, \ell) - \lambda_a - \tilde{\chi}$, then

$$\begin{aligned}\tau^a &= \frac{B - \bar{B}}{\zeta_\tau^{\text{HtM}}} \left(-\theta_t - \frac{\rho - r}{B} \iint a \left[\phi u_c(c, \ell) + (u_c(c, \ell) - \tilde{\chi})g \right] dadz + \mu + \zeta_B^{\text{HtM}} \right) \\ \tau^a &= \frac{B - \bar{B}}{\zeta_\tau^{\text{HtM}}} \left(-\theta_t - \frac{\rho - r}{B} \iint a u_c(c, \ell) \left[1 + \Theta \right] g dadz + \mu + \zeta_B^{\text{HtM}} \right)\end{aligned}$$

where $\Theta = \frac{\phi}{g} - \frac{\tilde{\chi}}{u_c(c, \ell)}$.

A.3 Proof of Proposition 9

Because this proof mimics the one for Proposition 5 but extends the aggregate state space, all the derivations until using calculus of variation to derive the necessary conditions are exactly the same. Therefore, I start by recalling the differentiated Lagrangian

$$\begin{aligned}0 = L_\alpha &= \int_0^\infty e^{-\rho t} \left\{ \iiint \left\{ \left\{ \left[u_c(c, \ell)h_c + u_\ell(c, \ell)h_\ell \right] g + \left[u(c, \ell) + \mu a \right] h_g \right. \right. \right. \\ &\quad + \phi \left[u_c(c, \ell)h_c + u_\ell(c, \ell)h_\ell \right] + \left[\frac{d}{d\alpha} \mathcal{A}^*(0)V + \mathcal{A}^*h_V \right] \phi - \phi_t h_V \\ &\quad + \chi u_{cc}(c, \ell)h_c + \chi u_{c\ell}(c, \ell)h_\ell + \chi_a h_V \\ &\quad + \psi u_{\ell\ell}(c, \ell)h_\ell + \psi u_{\ell c}(c, \ell)h_c - \psi_a(1 - \tau)wzh_V - \psi_a \left((B - \bar{B})h_\tau + \tau^a h_B \right) wzV \\ &\quad - \psi_a(1 - \tau)h_w zV \\ &\quad + \left. \frac{d}{d\alpha} \mathcal{A}(0)\lambda g + \mathcal{A}\lambda h_g + \lambda_t h_g - \rho \lambda h_g \right\} \Big\} dadz \\ &\quad + \theta \left[h_r B + (r - \rho)h_B - \left((B - \bar{B})h_\tau + \tau^a h_B \right) wL - \tau w \left(\iint z(g h_\ell + \ell h_g) dadz \right) \right] + \theta_t h_B \\ &\quad + \mu \left[-h_K - h_B \right] \Big\} d\Gamma \Big\} dt\end{aligned}$$

where, and here comes the main difference, for smooth functions f, g , we have

$$\begin{aligned}\iiint \frac{d}{d\alpha} \mathcal{A}(0) f g dadz &= \iiint \left[ah_r - \left((B - \bar{B})h_\tau + \tau^a h_B \right) w\ell + (1 - \tau)h_w z\ell + (1 - \tau)wh_\ell - h_c \right] f_a g dadz d\Gamma \\ &\quad + \iiint \left[h_K \partial_K \mu_K(\Gamma) + h_B \partial_B \mu_K(\Gamma) + h_\tau \partial_\tau \mu_K(\Gamma) \right] f_K g dadz d\Gamma \\ &\quad + \iiint \left[h_B \partial_B \mu_B(\Gamma) + h_K \partial_K \mu_B(\Gamma) + h_\tau \partial_\tau \mu_B(\Gamma) \right] f_B g dadz d\Gamma.\end{aligned}$$

The last two lines of the latter expression show only the first-order conditions for

government debt and the debt-adjustment coefficient. The two conditions that change in a meaningful way (that is, something extra apart from including an extra variable in the state-space) are:

Government debt with aggregate shocks. Regrouping the terms that are multiplied by h_B

$$0 = -\tau^a w \iint \ell z (\phi V_a + \lambda_a g) dadz - \tau^a w \iint \psi_a z V dadz - \theta \tau^a w L + \theta(r - \rho) - \theta_t - \mu + \mathcal{E}_B$$

where

$$\begin{aligned} \mathcal{E}_B(\Gamma) = & \iint \partial_B \mu_B(\Gamma) \phi V_B dadz + \iint \partial_B \mu_K(\Gamma) \phi V_K dadz \\ & + \iint \partial_B \mu_B(\Gamma) \lambda_B g dadz + \iint \partial_B \mu_K(\Gamma) \lambda_K g dadz \end{aligned} \quad (34)$$

and finally,

$$\theta_t = \tau^a w \iint \left[\ell z (\phi V_a + \lambda_a g) + \psi_a z V \right] dadz + \tau^a \theta w L + \theta(\rho - r) + \mu - \mathcal{E}_B$$

Debt-adjustment coefficient with aggregate shocks. Regrouping the terms multiplied by h_τ yields

$$0 = \int w(B - \bar{B}) \iint \left[\ell z (\phi V_a + \lambda_a g) + \psi_a z V \right] dadz d\Gamma + \int \theta(B - \bar{B}) w L d\Gamma + \mathcal{E}_\tau$$

where

$$\begin{aligned} \mathcal{E}_\tau = & \iiint \partial_\tau \mu_B(\Gamma) \phi V_B dadz d\Gamma + \iiint \partial_\tau \mu_K(\Gamma) \phi V_K dadz d\Gamma \\ & + \iiint \partial_\tau \mu_B(\Gamma) \lambda_B g dadz d\Gamma + \iiint \partial_\tau \mu_K(\Gamma) \lambda_K g dadz d\Gamma \end{aligned} \quad (35)$$

Accounting for boundary conditions, a “discretize-optimize” approach. Now (a, z, Γ) are discretized in a matrix in a scattered way, so that row i of the grid corresponds to an individual with wealth a_i , productivity z_i , in a world with aggregate state Γ_i . The grid for $\bar{\Gamma}$ has size I , and step sizes da , dz and $d\Gamma$. I denote for convenience $dx = dadz d\Gamma$. Many different i lead to the same aggregate state, I introduce a mapping $\phi(i)$ that maps each row i to a subset $\{1, \dots, J\}$ of aggregate states so that there is no $j \neq j'$ such that $\Gamma_j = \Gamma_{j'}$. I denote the inverse of this function by $\Phi(j)$ and it equals all the i 's that have aggregate

state j associated with them. Finally, $\Phi(j|H)$ is the subset of i 's associated with Γ_j that is hand-to-mouth, and its complement, the subset of unconstrained households, is denoted $\Phi(j|R)$. I assume that the first $\bar{I}$ entries of the grid correspond to hand-to-mouth agents, and the rest to unconstrained. The time grid, denoted $\{t_n\}_n$, has size N and time step dt . Indexing a variable $x_t(a, z, \Gamma)$ by x_{in} , the Lagrangian becomes

$$\begin{aligned}
L(g_0) = \sum_{n=0}^{N-1} e^{-\rho t_n} \Bigg\{ & \sum_i u(c_{in}) g_{in} dx - \sum_{i>\bar{I}} \frac{\phi_{in} - \phi_{in-1}}{dt} V_{in} dx + \sum_i \phi_{in} \left[u(c_{in}, \ell_{in}) + \mathcal{A}_{in} V_{in} \right] dx \\
& + \sum_{i>\bar{I}} \chi_{in} \left[u_c(c_{in}, \ell_{in}) - \left(\partial_a V \right)_{in} \right] dx \\
& + \sum_{i>\bar{I}} \psi_{in} \left[u_\ell(c_{in}, \ell_{in}) - (1 - \tau_n) w_{\phi(i)n} z_i \left(\partial_a V \right)_{in} \right] dx \\
& + \sum_{i>\bar{I}} \frac{\lambda_{in} - \lambda_{in-1}}{dt} g_{in} dx - \sum_{i>\bar{I}} \rho \lambda_{in} g_{in} dx + \sum_{i>\bar{I}} \lambda_{in} \mathcal{A}_{in}^* g_{in} dx \\
& - \sum_j \mu_{jn} \left[K_{jn} + B_{jn} \right] d\Gamma + \sum_i \mu_{\phi(i)n} a_i g_{in} dx \\
& + \sum_j \frac{\theta_{jn} - \theta_{jn-1}}{dt} B_{jn} d\Gamma \\
& + \sum_j \theta_{jn} \left[(r_{jn} - \rho) B_{jn} + G_{jn} - \tau_n w_{jn} L \right] d\Gamma \Bigg\} dt.
\end{aligned}$$

Interest Rate. We have

$$0 = \iint a \left[\phi V_a + \lambda_a g \right] dadz + \zeta_r^{\text{HtM}} + \theta B$$

where

$$\zeta_r^{\text{HtM}} = \left(\underline{a} + (1 - \tau) w z \underline{\ell}_r \right) u_c(\underline{c}, \underline{\ell}) \underline{g} dadz + (1 - \tau) w z \underline{\ell}_r u_\ell(\underline{c}, \underline{\ell}) \underline{g} dadz \quad (36)$$

where $\underline{c}$ (resp. $\underline{\ell}$ with derivative with respect to r denoted $\underline{\ell}_r$) denote consumption (resp. labor supply) of the constrained individual, and where $\underline{g} dadz$ is the measure of individuals at the Dirac mass point $(\underline{a}, \underline{z})$.

Government debt. We have

$$\theta_t = \tau^a w \iint z \left[\ell(\phi V_a + \lambda_a g) + \psi_a V \right] dadz + \tau^a \theta w L + \theta(\rho - r) + \mu + \zeta_B^{\text{HtM}} - \mathcal{E}_B$$

where

$$\zeta_B^{\text{HtM}} = \left(\tau^a w \underline{z} \underline{\ell} - (1 - \tau) w \underline{z} \underline{\ell}_B \right) u_c(\underline{c}, \underline{\ell}) \underline{g} d\alpha dz - (1 - \tau) w \underline{z} \underline{\ell}_B u_\ell(\underline{c}, \underline{\ell}) \underline{g} d\alpha dz \quad (37)$$

Debt-adjustment coefficient. And finally, we have

$$0 = \int w(B - \bar{B}) \iint z \left[\ell(\phi V_a + \lambda_a g) + \psi_a V \right] d\alpha dz d\Gamma + \int \theta(B - \bar{B}) w L d\Gamma + \zeta_\tau^{\text{HtM}} + \mathcal{E}_\tau$$

where

$$\zeta_\tau^{\text{HtM}} = \int \left((B - \bar{B}) w \underline{z} \underline{\ell} - (1 - \tau) w \underline{z} \underline{\ell}_\tau \right) u_c(\underline{c}, \underline{\ell}) \underline{g} d\alpha dz d\Gamma - \int (1 - \tau) w \underline{z} \underline{\ell}_\tau u_\ell(\underline{c}, \underline{\ell}) \underline{g} d\alpha dz d\Gamma \quad (38)$$

which concludes the proof.

A.4 Proof of Proposition 10 and Corollaries 12, 11 and 13

Using

$$\begin{aligned} \theta_t &= \tau^a w \iint z \left[\ell(\phi V_a + \lambda_a g) + \psi_a V \right] d\alpha dz + \tau^a \theta w L + \theta(\rho - r) + \mu + \zeta_B^{\text{HtM}} - \mathcal{E}_B \\ 0 &= \int (B - \bar{B}) w \left[\iint z \left[\ell(\phi V_a + \lambda_a g) + \psi_a V \right] d\alpha dz + \theta L \right] d\Gamma + \zeta_\tau^{\text{HtM}} + \mathcal{E}_\tau \end{aligned}$$

where the first line rewrites as

$$\frac{1}{\tau^a} \left(\theta_t - \theta(\rho - r) - \mu - \zeta_B^{\text{HtM}} + \mathcal{E}_B \right) = w \left[\iint z \left[\ell(\phi V_a + \lambda_a g) + \psi_a V \right] d\alpha dz + \theta L \right]$$

we get

$$0 = \int (B - \bar{B}) \frac{1}{\tau^a} \left(\theta_t - \theta(\rho - r) - \mu - \zeta_B^{\text{HtM}} + \mathcal{E}_B \right) d\Gamma + \zeta_\tau^{\text{HtM}} + \mathcal{E}_\tau$$

which yields

$$\tau^a = \frac{1}{\zeta_\tau^{\text{HtM}} + \mathcal{E}_\tau} \int (B - \bar{B}) \left(-\theta_t + \theta(\rho - r) + \mu + \zeta_B^{\text{HtM}} - \mathcal{E}_B \right) d\Gamma.$$

Using

$$\theta = -\frac{1}{B} \iint au_c(c, \ell) [1 + \Theta] g dadz$$

we finally get

$$\tau^a = \frac{1}{\zeta_\tau^{\text{HtM}} + \mathcal{E}_\tau} \int (B - \bar{B}) \left(-\theta_t - \frac{\rho - r}{B} \iint au_c(c, \ell) [1 + \Theta] g dadz + \mu + \zeta_B^{\text{HtM}} - \mathcal{E}_B \right) d\Gamma,$$

where in the long-run solution, $\theta_t = 0$. In the main text, I use the decomposition

$$\frac{1}{g_\Gamma(\Gamma)} \iint au_c(c, \ell) [1 + \Theta] g dadz = \mathbb{E}_{g|\Gamma} \left[au_c(c, \ell) [1 + \Theta] \right],$$

where $g_\Gamma(\Gamma) = \iint g(a, z, \Gamma) dadz$ and the expectation is taken over the pdf g conditional on an aggregate state Γ . Then, I decompose

$$\mathbb{E}_{g|\Gamma} \left[au_c(c, \ell) [1 + \Theta] \right] = A \mathbb{E}_{g|\Gamma} \left[u_c(c, \ell) [1 + \Theta] \right] + \text{Cov}_{g|\Gamma} \left[a, u_c(c, \ell) [1 + \Theta] \right].$$

For Corollary 12, I set $\mathcal{E}_B = \mathcal{E}_\tau = 0$. For Corollary 11, I use $g(a, z|\Gamma) \rightarrow \delta(a - A)\delta(z - z^{RA})g_\Gamma(\Gamma)$, where z^{RA} is the single level of labor productivity in the RA case and $\delta(\cdot)$ is the delta function. As a consequence, the covariance in the latter expression equals zero and the expectation tends to $u_c(C, L)(1 + \Theta)$. Corollary 13 follows from setting $\theta = 0$, which is, as discussed in [Dávila and Schaab \(2023\)](#), the continuous time counterpart of optimal discretion policy.

B Numerical Implementation

The algorithm has three steps: the deterministic steady state, the stochastic steady state for a given debt-adjustment coefficient, and the search for the optimal coefficient.

B.1 Step 1: Deterministic Steady State

The optimality conditions of the deterministic competitive equilibrium are

$$\begin{aligned}
\rho v(a, z) &= u(c, \ell) + \partial_a v(a, z) s(a, z) - \theta_z z \partial_z v(a, z) + \partial_{zz} v(a, z) \frac{\sigma_z^2}{2}; \quad a \geq \underline{a} \\
c &= (\partial_c u)^{-1}(\partial_a v) \ \& \ \ell = (\partial_\ell u)^{-1}\left(- (1 - \tau) w z \partial_a v\right) \\
0 &= \mathcal{A}^* g(a, z) \\
w &= (1 - \alpha) K^\alpha L^{-\alpha} \\
r &= \alpha K^{\alpha-1} L^{1-\alpha} - \delta \\
K + B &= \iint a g(a, z) da dz = A,
\end{aligned}$$

where $s(a, z) = (1 - \tau) w z \ell + r a - c$ is the optimal saving policy function and the HJB equation holds in the interior of the state space. As detailed in [Achdou et al. \(2022\)](#), this system of partial differential equations can be discretized as

$$\begin{aligned}
\rho \mathbf{v} &= u(\mathbf{c}, \mathbf{l}) + \mathbf{A} \mathbf{v}; \quad \mathbf{c} = (u_c)^{-1}(D_a \mathbf{v}) \ \& \ \mathbf{l} = (u_\ell)^{-1}\left(- (1 - \tau) w \mathbf{z} D_a \mathbf{v}\right) \\
0 &= \mathbf{A}' \mathbf{g} \\
\mathbf{p} &= [w, r] = F(\mathbf{g}),
\end{aligned}$$

where $\mathbf{v}$ is the discrete counterpart of v , and likewise for $\mathbf{c}$, $\mathbf{l}$, and $\mathbf{g}$. The finite-difference matrix D_a discretizes the derivative with respect to wealth a . The matrix $\mathbf{A}$ discretizes the infinitesimal generator, and its transpose discretizes the adjoint. The price vector $\mathbf{p}$ solves a fixed-point problem, summarized by $F(\cdot)$, that ensures market clearing for capital and labor. I use the deterministic steady state to calibrate the model as described in [Section 5.1](#) and to initialize the algorithm for the stochastic steady state.

B.2 Step 2: Stochastic Steady State

The optimality conditions of the stochastic competitive equilibrium are

$$\begin{aligned}
\rho v(a, z, \Gamma) &= u(c, \ell) + \partial_a v(a, z, \Gamma) s(a, z, \Gamma) - \theta_z z \partial_z v(a, z, \Gamma) + \partial_{zz} v(a, z, \Gamma) \frac{\sigma_z^2}{2} \\
&\quad + \partial_\Gamma v(a, z, \Gamma) \mu(\Gamma) + \partial_{ZZ} v(a, z, \Gamma) \frac{\sigma_z^2}{2}; \quad a \geq \underline{a}
\end{aligned}$$

$$\begin{aligned}
0 &= \mathcal{A}^* g(a, z, \Gamma) \\
w &= (1 - \alpha) Z K^\alpha L^{-\alpha} \\
r &= \alpha Z K^{\alpha-1} L^{1-\alpha} - \delta \\
K + B &= \iint ag(a, z | \Gamma) dadz = A \quad \forall \Gamma,
\end{aligned}$$

where $\mu(\Gamma)$ collects the drifts of the aggregate state, including the perceived laws of motion $\mu_K(\Gamma)$ and $\mu_B(\Gamma)$. The system is discretized as in Step 1, with the state space augmented by Γ .

The algorithm follows [Achdou et al. \(2022\)](#), extended to aggregate risk as in [Krusell and Smith \(1998\)](#) and [Fernández-Villaverde et al. \(2023\)](#), and it relies on the finite-difference method ([LeVeque, 2007](#)). To speed up computations, I use sparse grids with the tools developed by [Schaab and Zhang \(2022\)](#), instead of the dense grids used in [Achdou et al. \(2022\)](#) and [Fernández-Villaverde et al. \(2023\)](#).

Algorithm. (1) Choose perceived laws of motion parametrized by weights $(\beta_i^\ell)_i$. (2) Given μ , solve the HJB equation. (3) Simulate a realization of the TFP sequence $\{Z_t\}_t$. (4) Given the simulated path of TFP, solve the Kolmogorov forward equation to obtain $\{g_t(\cdot)\}_t$, and use it to compute aggregate capital $\{K_t\}_t$ and government debt $\{B_t\}_t$. (5) Regress, by ordinary least squares, the simulated changes $\{dK_t\}_t$ and $\{dB_t\}_t$ on the simulated aggregate state $\{\Gamma_t\}_t$,²³ and update the weights to $\beta^{\ell+1}$. If the weights have converged, stop; otherwise, return to step (2).

B.3 Step 3: Optimal Debt-Adjustment Coefficient

For a given coefficient τ^a , Step 2 delivers the stochastic steady state. I compute utilitarian welfare $\mathcal{W}$ by Monte Carlo simulation in the stochastic steady state. I then search over τ^a for the coefficient that maximizes $\mathcal{W}$, recomputing the stochastic steady state, and hence the perceived laws of motion, for each candidate coefficient, so that the implementability conditions hold at every point of the search.

²³ [Fernández-Villaverde et al. \(2023\)](#) use a neural network. Because approximate aggregation holds in this model, a linear perceived law of motion is sufficient.